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(54) **BAICALEIN- AND
SCUTELLAREIN-SYNTHESIZING
MICROORGANISM, PREPARATION
METHOD AND APPLICATIONS THEREOF**

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2319/10; **C07K 2319/20**; **C07K 2319/00**;
C12P 17/06; **C12Y 106/02004**; **C12Y**
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See application file for complete search history.

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(57) **ABSTRACT**

Provided are a baicalein- and scutellarein-synthesizing microorganism, a preparation method for same, and applications thereof. By modifying a heterologous metabolic pathway of a host cell per a genetic engineering method, acquired is an engineered strain providing a high yield of baicalein and scutellarein. Also provided is a process for utilizing the engineered strain to produce baicalein and scutellarein.

7 Claims, 7 Drawing Sheets

Specification includes a Sequence Listing.

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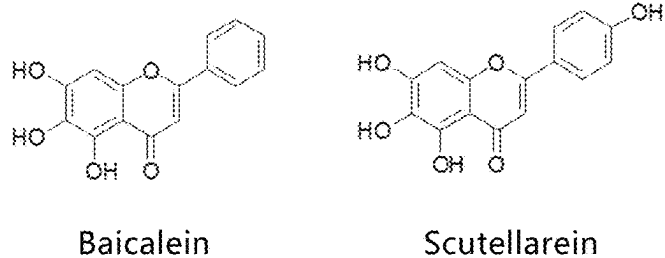


Figure 1

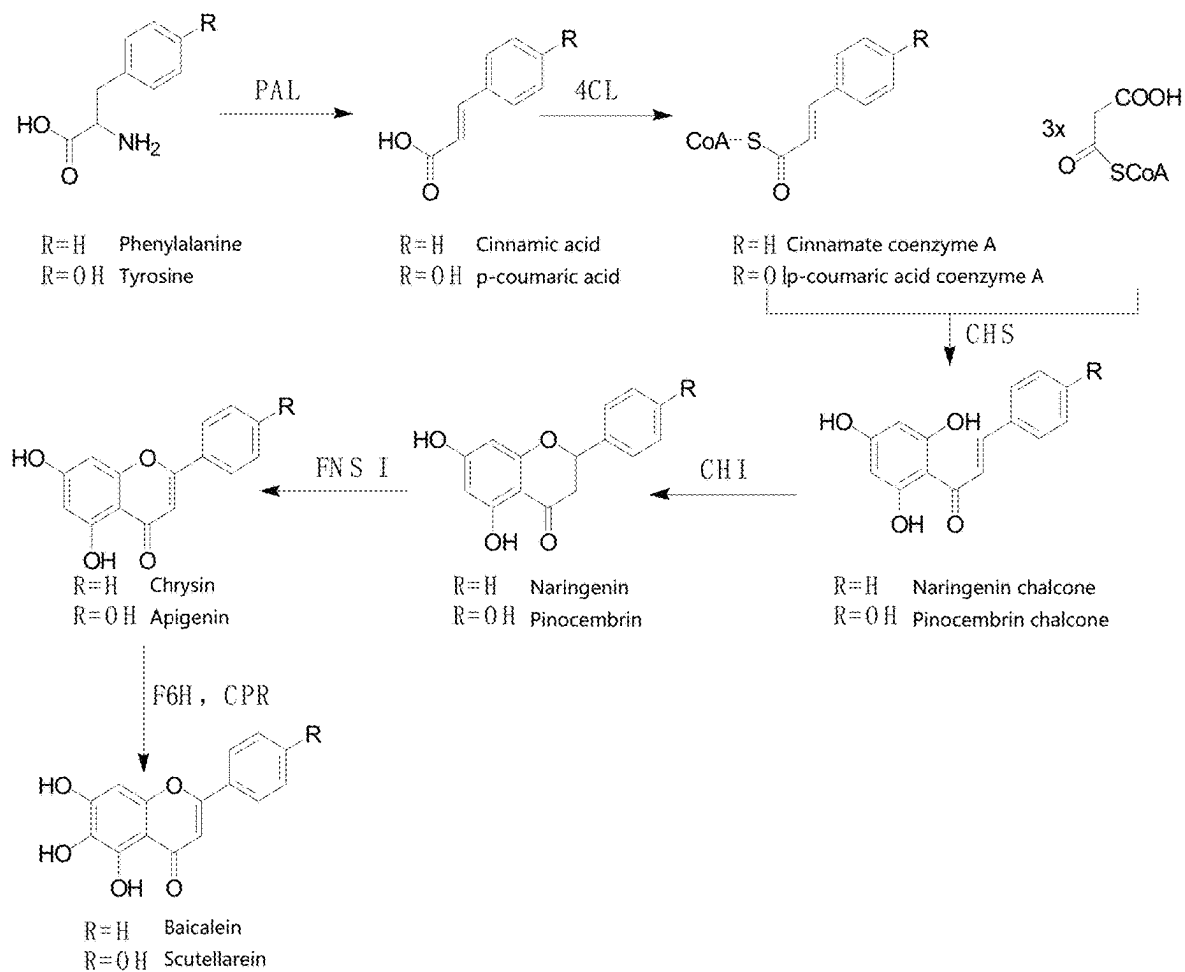


Figure 2

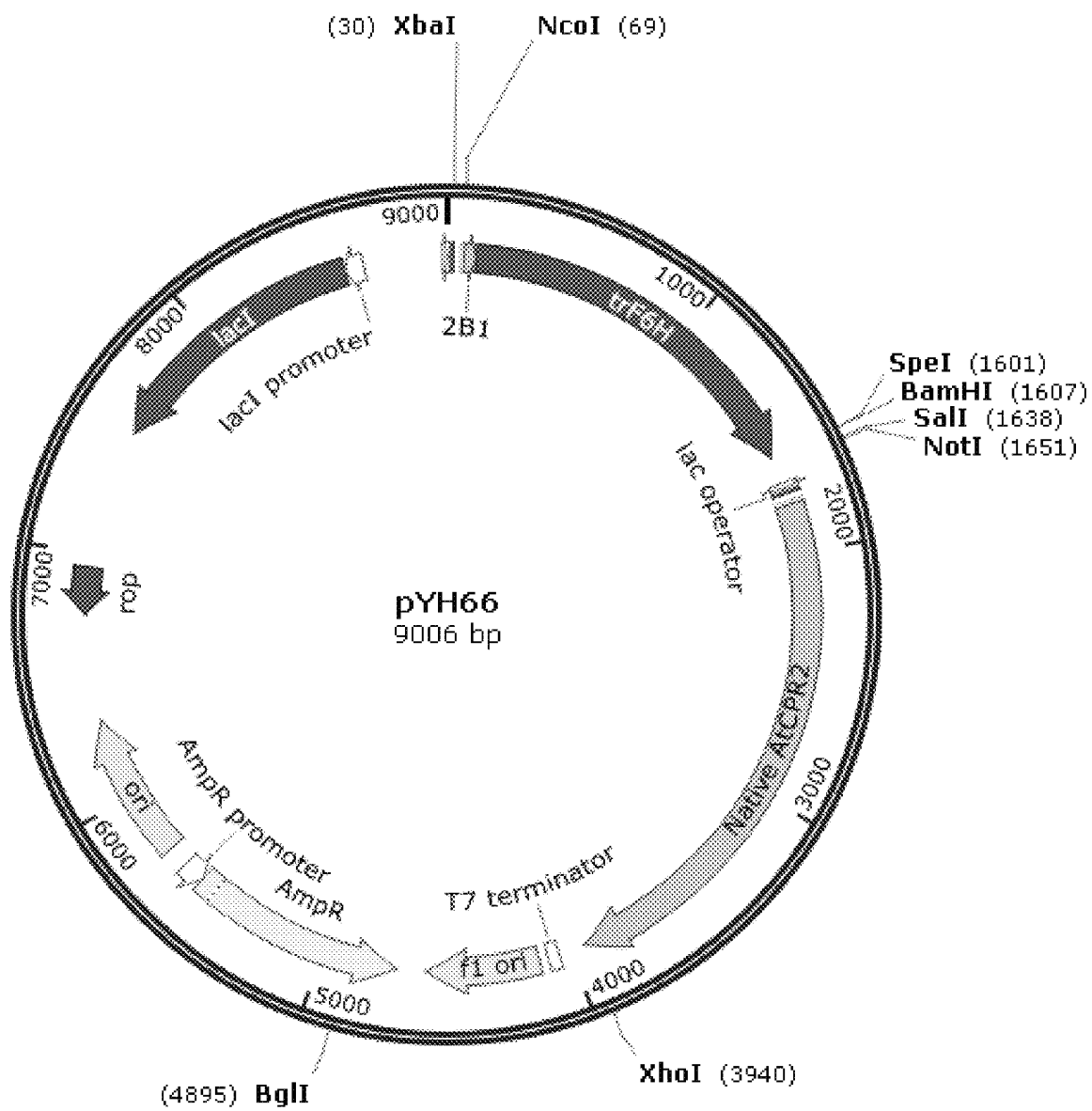


Figure 3

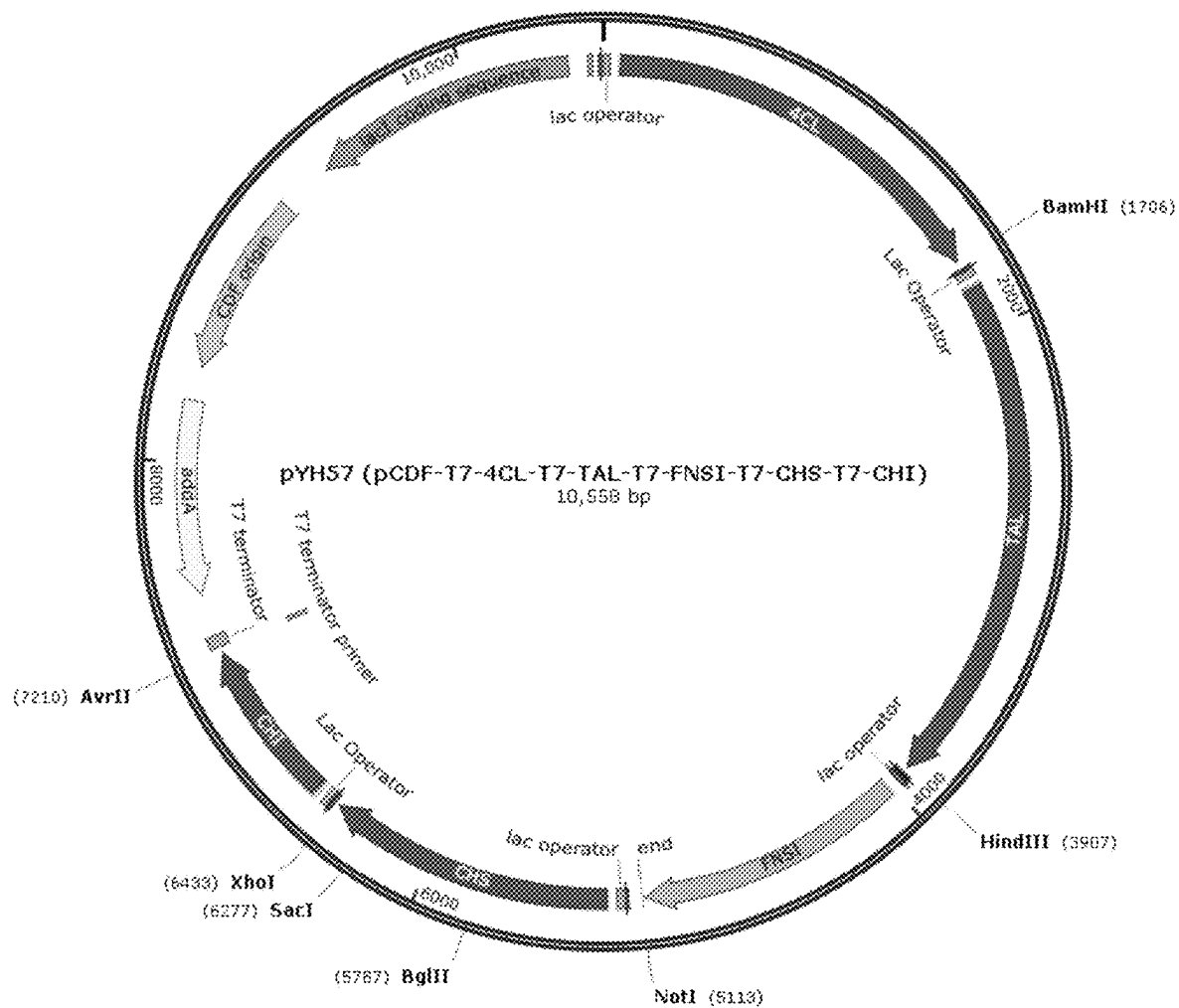


Figure 4

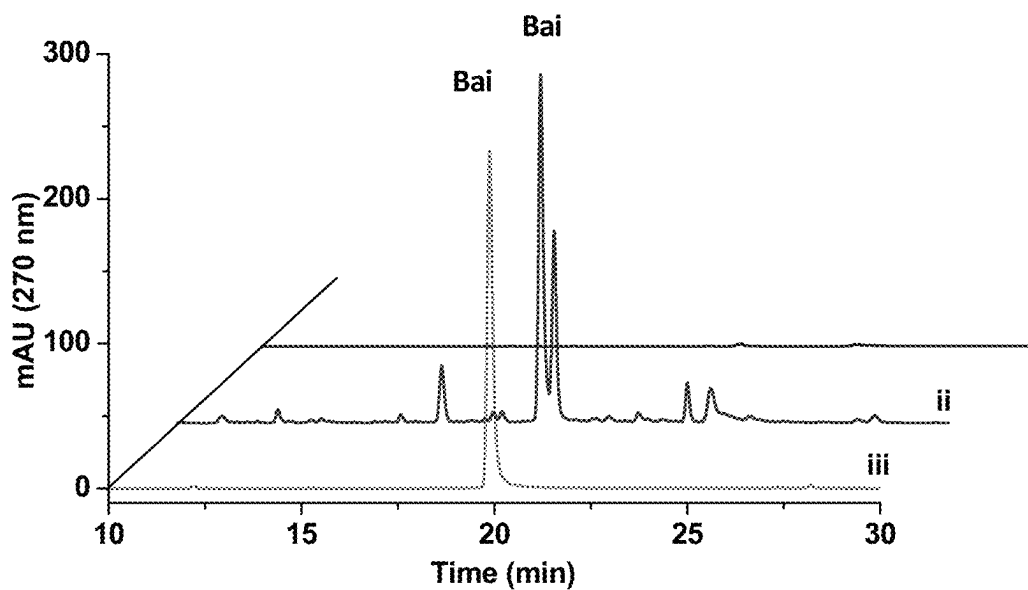


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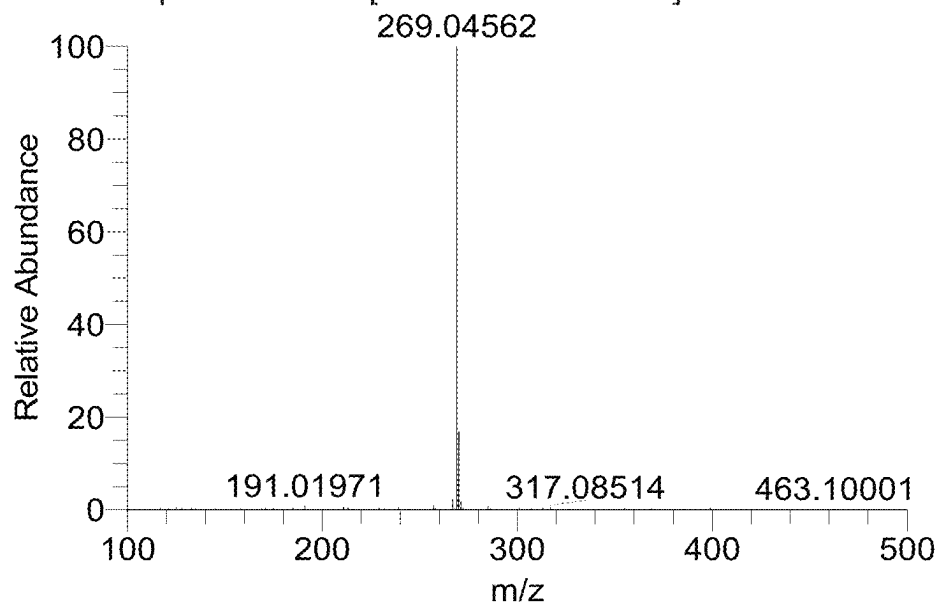


Figure 6

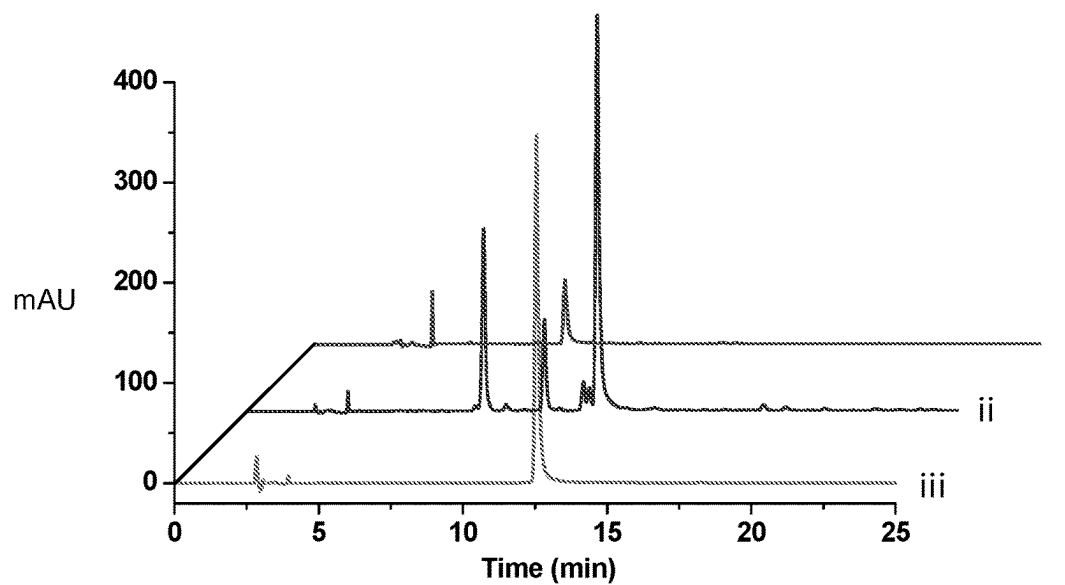


Figure 7

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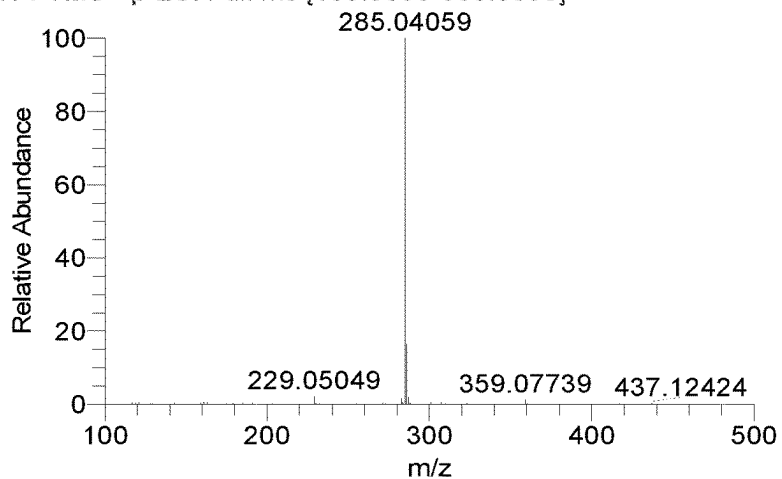


Figure 8

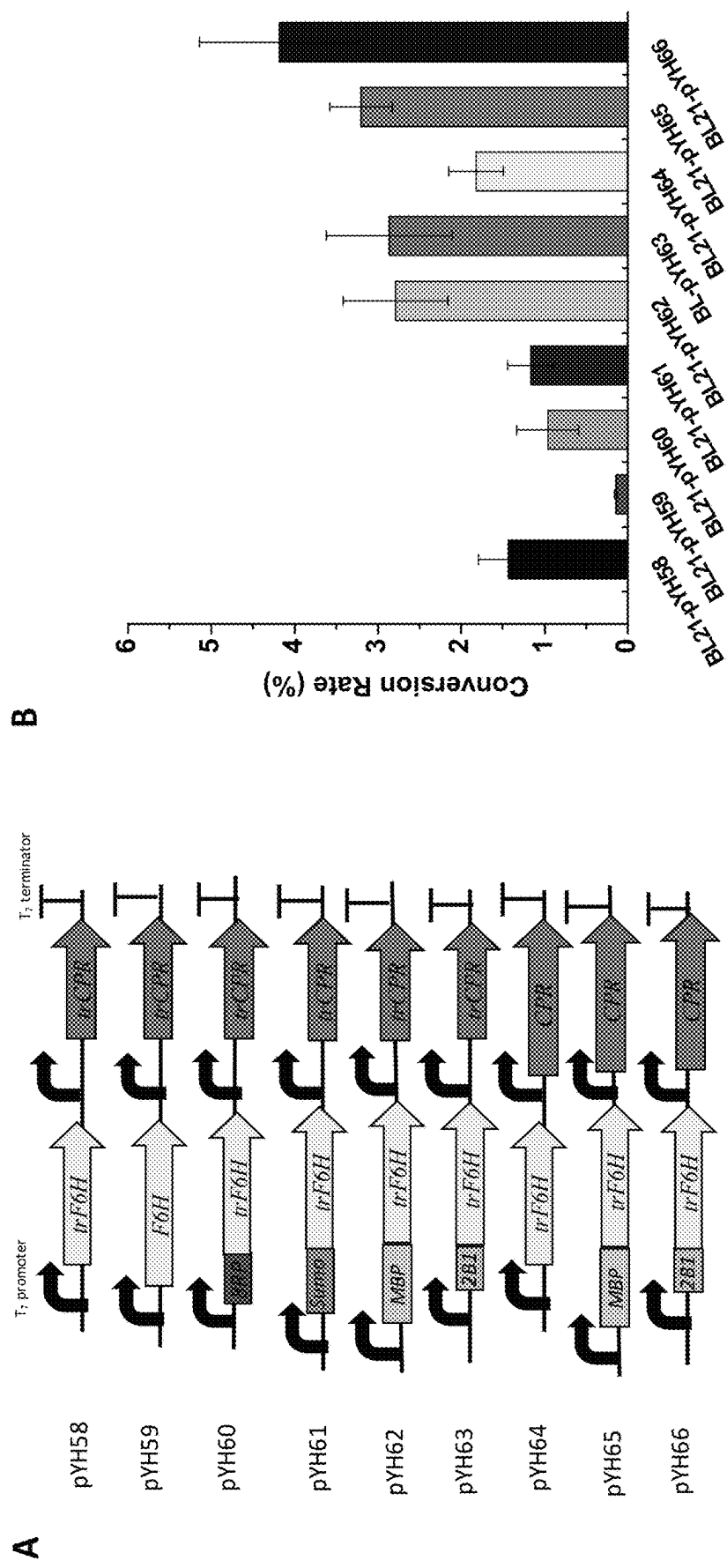


Figure 9

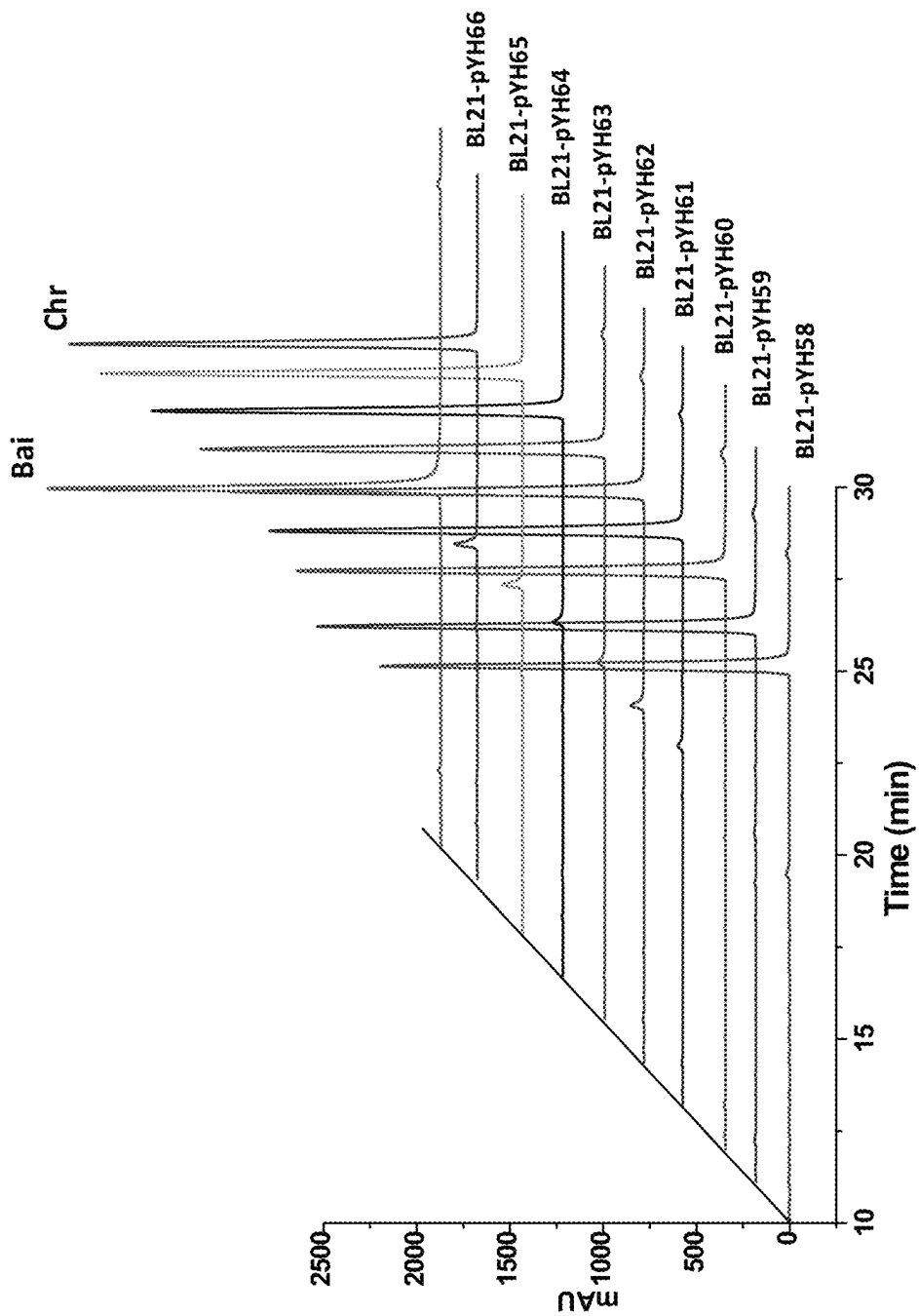


Figure 9, continued

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BAICALEIN- AND SCUTELLAREIN-SYNTHESIZING MICROORGANISM, PREPARATION METHOD AND APPLICATIONS THEREOF

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Phase application under 35 U.S.C. 371 of International Application No. PCT/CN2019/104658 filed Sep. 6, 2019, which claims priority to and the benefit of Chinese Patent Application No. 201811043657.0 filed Sep. 7, 2018, the entire disclosure of each of which is incorporated herein by reference.

FIELD

The invention relates to the technical fields of synthetic biology and medicine, in particular to a microorganism for synthesizing baicalein and scutellarein, a preparation method and application thereof.

BACKGROUND

Scutellaria baicalensis Georgi is a famous traditional medicine in China, which is a *Labiatae* plant. Traditional Chinese medicine (TCM) Radix *Scutellariae* is the dry root of *Scutellaria baicalensis* Georgi, which has a long medicinal history and can be used for the treatment of wind heat, damp heat and other diseases. *Erigeron* Herba is the dry herb of *Erigeron breviscapus*. It is cold in nature and bitter in taste. It has the functions of anti-inflammatory and analgesic, promoting blood circulation and removing blood stasis, eliminating wind and dampness. Extract of *Scutellaria baicalensis* Georgi and *Erigeron breviscapus* have long been widely used in TCM preparations. The main raw materials of Yinhuang tablet, Shuanghuanglian oral liquid and Lanqin oral liquid are extract of *Scutellaria baicalensis* Georgi. The main active ingredient of Qingkailing is baicalein, which has the effect of anti-inflammatory, prevention and treatment of diarrhea, liver disease and tumor. The common dosage forms of *Erigeron breviscapus* include Breviscapine tablet and Breviscapine oral liquid, which can be converted to scutellarein and absorbed by an organism. Therefore, baicalein and scutellarein have a certain value in the development of new drugs.

Baicalein and scutellarein are two important flavonoids (flavones) with similar structures. The molecular formula of baicalein is $C_{15}H_{10}O_5$ with a molecular weight of 270.24, while molecular formula of scutellarein is $C_{15}H_{10}O_6$ with a molecular weight of 286.24. Their structures are shown in FIG. 1.

Like most natural products from plants, baicalein and scutellarein are mainly prepared by chemical synthesis and organic solvent extraction. Organic solvent extraction extracts ingredient from the tissues of *Scutellaria baicalensis* Georgi, *Erigeron breviscapus*, *Scutellaria barbata* and other medicinal plants, which needs a lot of organic solvents and complex separation process. Therefore, the cost of organic solvent extraction is high. In addition, the main problems are that plants grow slow and medicinal resources are destroyed. Although baicalein and scutellarein can also be obtained in large quantities through chemical synthesis, the raw materials in the synthesis process contain cinnamic acid or its derivatives, oxyphenol and other chemical substances, which to some extent limits its application in

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medicine and food field. In addition, toxic reagents and expensive chemical catalysts are also used in the synthesis process.

Synthetic biology is a discipline which integrates and assembles standardized biological components based on rational design to build an excellent artificial life system. As soon as synthetic biology was put forward, its ideas and design have a profound impact on the development of industrial microbial technology, which makes microbial technology play a greater role in the development and production of drugs, biofuels and fine chemicals.

In the art, the synthetic elements of various natural products are assembled to conduct heterologous synthesis in microorganisms. However, the two flavonoids baicalein and scutellarein have not been successfully heterologously synthesized in microorganisms. Therefore, it is urgent to construct a microbial strain capable of heterologous synthesis of baicalein and scutellarein.

SUMMARY

Provided are a baicalein- and scutellarein-synthesizing microorganism, a preparation method for same, and applications thereof.

The first aspect of the present disclosure provides a method of producing baicalein and scutellarein, comprising: (1) introducing into a host cell genes expressing flavone 6-hydroxylase (F6H) and cytochrome P450 oxidoreductase (CPR), as well as genes for synthesizing chrysin or apigenin; and (2) culturing the host cell in a culture system containing phenylalanine and/or tyrosine to produce baicalein or scutellarein.

In a preferable example, the genes for synthesizing chrysin or apigenin comprises: genes expressing phenylalanine ammonia-lyase (PAL), 4-coumarate: CoA ligase (4CL), chalcone synthase (CHS), chalcone isomerase (CHI) and flavone synthase I (FNSI). Preferably, when introduced into the host cell, the genes expressing PAL, 4CL, CHS, CHI and FNSI are in the same expression vector.

In another preferable example, the flavone 6-hydroxylase is derived from *Scutellaria baicalensis*, including its homologues (homologous genes or peptides from other species); the CPR is derived from *Arabidopsis thaliana*, including its homologues.

In another preferable example, the PAL is derived from *Rhodotorula toruloides*, including its homologues; the 4CL is derived from *Petroselinum crispum*, including its homologues; the CHS is derived from *Petunia X hybrida*, including its homologues; the CHI is derived from *Medicago sativa*, including its homologues; and the FNS I is derived from *Petroselinum crispum*, including its homologues.

The another aspect of the present disclosure provides a method of producing baicalein and scutellarein, comprising: (1) introducing into a host cell genes expressing flavone 6-hydroxylase (F6H) and cytochrome P450 oxidoreductase (CPR) to obtain recombinant host cell; and (2) culturing the recombinant host cell in a culture system containing chrysin or apigenin to produce baicalein or scutellarein.

In another aspect of the disclosure, a method for converting chrysin or apigenin into baicalein or scutellarein is provided: catalyzing chrysin or apigenin by flavone 6-hydroxylase and cytochrome P450 oxidoreductase, thereby adding a hydroxyl group to the structure of chrysin or apigenin to form baicalein or scutellarein.

In a preferable embodiment, the flavone 6-hydroxylase (F6H) is a mutant flavone 6-hydroxylase with the N-terminal amino acids (1-10) to (20-30) truncated; preferably, it is a

mutant flavone 6-hydroxylase with the N-terminal amino acids (2-5) to (22-28) truncated.

In another preferable embodiment, the flavone 6-hydroxylase is fused with a peptide tag, and the peptide tag is selected from N-terminal 8 amino acid peptide of bovine calf serum 17 hydroxylase (8RP), small ubiquitin-related modifier (Sumo), maltose binding protein (MBP), 2B1 family soluble protein of cytochrome P450 (2B1), or a combination thereof, preferably the peptide tag is maltose binding protein or 2B1 family soluble protein of cytochrome P450, or a combination thereof; preferably, the peptide tag is located at the N-terminal.

In another preferable embodiment, the cytochrome P450 oxidoreductase (CPR) is a mutant cytochrome P450 oxidoreductase with the N-terminal amino acids (1-20) to (60-85) truncated; preferably, it is a mutant cytochrome P450 oxidoreductase with the N-terminal amino acids (2-10) to (65-80) truncated; more preferably, it is a mutant cytochrome P450 oxidoreductase with the N-terminal amino acids (2-5)-(70-75) truncated.

In another preferable embodiment, the host cell includes: prokaryotic cell or eukaryotic cell; preferably, the prokaryotic cell includes: *Escherichia coli* cell, *Bacillus subtilis* cell; the eukaryotic cell includes: yeast cell.

Another aspect of the present disclosure provides a recombinant host cell comprising exogenous genes expressing flavone 6-hydroxylase and cytochrome P450 oxidoreductase.

In another preferable embodiment, the recombinant host cell also includes exogenous genes for synthesizing chrysin or apigenin.

In another preferable embodiment, the peptide tag is a single copy or 2-10 copies (such as 3, 4, 5, 6, 8 copies) of tandem sequences.

Another aspect of the present disclosure provides the use of any of the above recombinant host cells in the production of baicalein and scutellarein.

In one preferable embodiment, for the strain which does not comprise chrysin or apigenin synthesis gene(s) in the cell, the use is to produce baicalein and scutellarein with exogenous chrysin or apigenin as the substrate; for the strain which comprises chrysin or apigenin synthesis gene(s) in the cell, the use is to produce baicalein and scutellarein in the presence of exogenous phenylalanine and/or tyrosine.

Another aspect of the disclosure provides a method of preparing a host cell for producing baicalein and scutellarein, comprising: introducing genes expressing flavone 6-hydroxylase and cytochrome P450 oxidoreductase into the host cell to obtain a recombinant strain; preferably, the method also comprises: introducing genes for synthesizing chrysin or apigenin.

In another aspect of the present disclosure, a kit for the production of baicalein and scutellarein is provided, wherein the kit comprises any of the above recombinant host cells.

In another preferable embodiment, the kit also comprises: culture medium for the host cell, instruction for use, etc.

In another aspect of the present disclosure, a mutant flavonoid 6-hydroxylase is provided, which corresponds to the wild-type flavonoid 6-hydroxylase (F6H) but the N-terminal amino acids (1-10) to (20-30) are truncated; preferably, the N-terminal amino acids (2-5) to (22-28) are truncated; more preferably, the mutant flavonoid 6-hydroxylase has the amino acid sequence shown in SEQ ID NO: 2.

In another aspect of the present disclosure, a mutant cytochrome P450 oxidoreductase is provided, which corresponds to the wild-type cytochrome P450 oxidoreductase but the N-terminal amino acids (1-20) to (60-85) are truncated;

preferably, the N-terminal amino acids (2-10) to (65-80) are truncated; more preferably, the N-terminal amino acids (2-5) to (70-75) are truncated; preferably, the mutant cytochrome P450 oxidoreductase has the amino acid sequence shown in SEQ ID NO: 8.

In another aspect of the present disclosure, a fusion polypeptide is provided, which comprises any of the above mutant flavone 6-hydroxylase fused with a peptide tag, the peptide tag is selected from the group consisting of: 8RP, Sumo, MBP, 2B1; preferably is MBP or 2B1.

In a preferable embodiment, the fusion polypeptide has an amino acid sequence selected from the group consisting of: SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5 or SEQ ID NO: 6.

In another aspect of the present disclosure, a polynucleotide is provided, which encodes: the mutant flavonoid 6-hydroxylase; or the mutant cytochrome P450 oxidoreductase; or the fusion polypeptide.

In another aspect of the present disclosure, an expression construct is provided, which comprises: any of the above polynucleotides; or polynucleotides encoding any of the mutant flavonoid 6-hydroxylase or the fusion protein described above, and polynucleotides encoding the mutant cytochrome P450 oxidoreductase described above.

In another preferable embodiment, the expression construct also comprises promoter and terminator operably linked with the above polynucleotide.

Another aspect of the disclosure provides the use of the mutant flavonoid 6-hydroxylase or the fusion protein and the mutant cytochrome P450 oxidoreductase in production of baicalein or scutellarein by adding a hydroxyl group to the structure of chrysin or apigenin.

Other aspects of the disclosure will be apparent to those skilled in the art based on the disclosure herein.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1. Structural formula of baicalein and scutellarein.

FIG. 2. Schematic of biosynthesis pathway of baicalein and scutellarein.

FIG. 3. Schematic of plasmid pYH66.

FIG. 4. Schematic of plasmid pYH57.

FIG. 5. HPLC results of engineering strain BL21(DE3)-pYH57-pYH66 and baicalein standard. i: BL21(DE3)-pETDuet-1-pCDFDuet-1 fermentation broth, as blank control; ii: BL21(DE3)-pYH57-pYH66 fermentation broth added with phenylalanine; baicalein standard.

FIG. 6. Mass spectrum results of baicalein produced by engineering strain BL21(DE3)-pYH57-pYH66.

FIG. 7. HPLC results of engineering strain BL21(DE3)-pYH57-pYH66 and scutellarein standard. i: BL21(DE3)-pETDuet-1-pCDFDuet-1 fermentation broth, as blank control; ii: BL21(DE3)-pYH57-pYH66 fermentation broth added with tyrosine; iii: scutellarein standard.

FIG. 8. Mass spectrum results of scutellarein produced by engineering strain BL21(DE3)-pYH57-pYH66.

FIG. 9. Production of baicalein from chrysin catalyzed by SbF6H and AtCPR mutants.

A. Schematic of the key elements in the constructed plasmid;

B. The conversion rates of baicalein from chrysin in recombinant *E. coli*;

C. HPLC results of the catalytic reaction solution of recombinant *E. coli*. Chr: chrysin; Bai: baicalein.

DETAILED DESCRIPTION

The inventor is committed to the heterologous synthesis of baicalein and scutellarein from microorganisms, and to

improving biological production of baicalein and scutellarein. After in-depth study, engineering strains with high yield of baicalein and scutellarein is obtained by modifying the heterologous metabolic pathway of host cells through genetic engineering.

As used herein, "N-terminal amino acids (1-10) to (20-30)" refers to a sequence starting from any amino acid in N-terminal amino acids 1-10 and ending at any amino acid in N-terminal amino acids 20-30.

As used herein, "N-terminal amino acids (2-5) to (22-28)" refers to a sequence starting from any amino acid in N-terminal amino acids 2-5 and ending at any amino acid in N-terminal amino acids 22-28.

As used herein, "N-terminal amino acids (1-20) to (60-85)" refers to a sequence starting from any amino acid in N-terminal amino acids 1-20 and ending at any amino acid in N-terminal amino acids 60-85.

As used herein, "N-terminal amino acids (2-10) to (65-80)" refers to a sequence starting from any amino acid in N-terminal amino acids 2-10 and ending at any amino acid in N-terminal amino acids 65-80.

As used herein, "N-terminal amino acids (2-5) to (70-75)" refers to a sequence starting from any amino acid in N-terminal amino acids 2-5 and ending at any amino acid in N-terminal amino acids 70-75.

As used herein, "exogenous" or "heterologous" refers to two or more nucleic acid or protein sequences from different sources.

As used herein, "operably linked (to)" or "operably connected (to)" is intended to mean a functional spatial arrangement between two or more nucleic acid regions or nucleic acid sequences. For example, a promoter region is "operatively linked" to the nucleic acid sequence of a target gene when the promoter region is placed at a specific position relative to the nucleic acid sequence so that the transcription of the nucleic acid sequence is guided by the promoter region.

As used herein, the "expression construct" refers to a recombinant DNA molecule that contains the desired nucleic acid coding sequence. An expression construct may contain one or more gene expression cassettes. The "construct" is usually contained in an expression vector.

As used herein, the PAL, 4CL, CHS, CHI and FNSI proteins are proteins that constitute the biosynthesis pathway of chrysin or apigenin in the expression system.

As used herein, the F6H and CPR proteins are the proteins that convert chrysin or apigenin into baicalein or scutellarein in the expression system.

Wild types of the above proteins or genes have been identified in the art, so they can be available and prepared from the public. As a preferable embodiment of the disclosure, PAL is derived from *Rhodotorula toruloides*, with the sequence shown in GenBank accession number AAA33883.1; 4CL is derived from *Petroselinum crispum*, with the sequence shown in GenBank accession number KF765780.1; CHS is derived from *Petunia X hybrida*, with the sequence shown in GenBank accession number KF765781.1; CHI is derived from *Medicago sativa*, with the sequence shown in GenBank accession number KF765782.1; FNS I is derived from *Petroselinum crispum*, with the sequence shown in Swiss-Prot accession number Q7XZQ8.1.

Wild types of F6H and CPR have also been identified in the art. As a preferable embodiment of the disclosure, F6H is derived from *Scutellaria baicalensis*, with the sequence shown in GenBank accession number ASW21050.1. As a preferable embodiment of the disclosure, CPR is derived

from *Arabidopsis thaliana*, with the sequence shown in GenBank accession number NP_849472.2.

The inventor found that when using host cells to produce baicalein and scutellarein, the wild-type F6H can only produce a small amount of products, which cannot achieve large-scale production. Through modification of multiple proteins involved in the reaction and a large number of screening and analysis, optimized modification schemes were obtained, which greatly improved the yield of baicalein and scutellarein of microorganisms, especially prokaryotic expression systems such as *E. coli*.

Therefore, a preferable embodiment of the present disclosure provides a mutant F6H that corresponds to the wild-type F6H with N-terminal amino acids (1-10) to (20-30) truncated; preferably, it is a mutant F6H with N-terminal amino acids (2-5) to (22-28) truncated; more preferably, it is a mutant F6H with N-terminal amino acids 2 to 25 truncated.

In a preferable embodiment of the disclosure, a fusion protein containing F6H or mutant F6H is provided, which includes F6H or any mutant F6H, and a peptide tag fused therewith, wherein the peptide tag is selected from the group consisting of 8RP, Sumo, MBP, 2B1, or a combination of them; preferably is MBP or 2B1. The peptide tag and the F6H or mutant F6H may or may not contain a linker peptide, and the linker peptide does not affect their biological activities.

In a preferable embodiment of the present disclosure, a mutant CPR is provided, which corresponds to the wild-type CPR with N-terminal amino acids (1-20) to (60-85) truncated; preferably, it is a mutant CPR with N-terminal amino acids (2-10) to (65-80) truncated; more preferably, it is a mutant CPR with N-terminal amino acids (2-5) to (70-75) truncated.

In addition to the above preferable proteins (including the above wild-type proteins and mutant proteins), the disclosure also includes their bioactive fragments, derivatives and analogues. Their fragments, derivatives or analogues may comprise deletion, insertion and/or substitution of several (usually 1-50, more preferably 1-20, yet more preferably 1-10, 1-5, 1-3, or 1-2) amino acids, as well as addition or deletion of one or more (for example, less than 100, 80, 50, 20, more preferably less than 10, yet more preferably less than 5) amino acids at C-terminal and/or N-terminal. For example, substitution with amino acids of comparable or similar properties usually does not change protein function in the art. As another example, addition or deletion of one or more amino acids to the C-terminus and/or N-terminus usually does not change the function of a protein either. However, for further variation of the above mutant protein, the N-terminal was truncated as described above.

In addition to the above preferable proteins (including the above wild-type proteins and mutant proteins), the disclosure also includes their analogues. The differences between analogs and the original protein may be the difference in amino acid sequences, and may also be the difference in the forms of modifications that will not affect the sequence, or both. These proteins include natural or induced genetic variants. Induced variants can be obtained by a variety of techniques, such as generating random mutagenesis by irradiation or exposure to mutagens, and can also be obtained by directed mutagenesis or other known molecular biology techniques. Analogs mentioned herein also include analogs with residue(s) different from natural L-amino acid (e.g., D-amino acids), as well as analogs with a non-naturally occurred or synthetic amino acid (such as β ,

γ -amino acids). It should be understood that the proteins of the present disclosure are not limited to the representative proteins described above.

In addition to the above preferable proteins (including the above wild-type proteins and mutant proteins), the disclosure also includes the protein with high homology (for example, having 70% or higher, preferable 80% or higher, more preferable 90% or higher (such as 95%, 98% or 99%) homology with the sequence of the particular described protein) and having the same function as the corresponding protein.

The disclosure describes proteins or genes from specific species. It should be understood that although the proteins or genes obtained from a specific species are preferably studied in the present disclosure, other proteins or genes obtained from other species and having high homology (such as having more than 60%, such as 70%, 80%, 85%, 90%, 95%, or even 98% sequence identity) with the proteins or genes also fall within the scope of the present disclosure.

The disclosure also provides a polynucleotide sequence encoding the protein of the disclosure or a conserved variant thereof. The polynucleotide sequences herein can be in the form of DNA or RNA. Forms of DNA include cDNA, genomic DNA or artificially synthesized DNA. DNA can be single-stranded or double-stranded. The DNA may be coding strand or non-coding strand. The polynucleotide encoding the mutant mature protein of the disclosure includes: the coding sequence only encoding the mature protein; the coding sequence encoding the mature protein and a various additional coding sequence; the coding sequence encoding the mature protein (and an optional additional coding sequence) and a noncoding sequence.

The disclosure also includes the codon-optimized polynucleotide sequence of the gene sequence, for example, the codon-optimized according to the codon bias of the host cell.

In the disclosure, an engineering strain with high yield of baicalein and scutellarein is also constructed, which includes exogenous genes expressing F6H (especially the mutant F6H or fusion protein) and CPR (especially the mutant CPR or fusion protein). Baicalein or scutellarein can be produced by culturing the recombinant strain and adding chrysin or apigenin into the culture system.

In the disclosure, another engineering strain with high yield of baicalein and scutellarein is constructed, which includes exogenous genes expressing F6H (especially the mutant F6H or fusion protein) and CPR (especially the mutant CPR or fusion protein), as well as genes for synthesizing chrysin or apigenin. The genes for synthesizing chrysin or apigenin comprise genes expressing PAL, 4CL, CHS, CHI and FNSI proteins.

By use of the strain according to the disclosure, which has great stability, large-scale cultivation and production of baicalein or scutellarein in a bioreactor can be realized. The yield of baicalein or scutellarein of the optimized strain of the disclosure is very high.

In the disclosure, more economical and convenient manufacture of baicalein or scutellarein can be conducted by production of baicalein or scutellarein from *E. coli*.

The disclosure also provides a kit for producing baicalein or scutellarein engineering strains. In addition, it can also include culture medium for *E. coli*, separation or detection reagent for baicalein or scutellarein, instruction for use, etc.

The disclosure is further illustrated by the specific examples described below. It should be understood that these examples are merely illustrative, and do not limit the scope of the present disclosure. The experimental methods without specifying the specific conditions in the following

examples generally used the conventional conditions, such as those described in J. Sambrook, Molecular Cloning: A Laboratory Manual (3rd ed. Science Press, 2002) or followed the manufacturer's recommendation.

5 Experimental Materials

AxyPrep Total RNA Miniprep Kit, PCR Gel Extraction Kit, Plasmid Extraction Kit are from Axygen; PrimeScript RT reagent Kit with gDNA Eraser (Perfect Real Time), PrimeSTAR Max DNA Polymerase are from Takara, and restriction enzymes are from NEB.

E. coli DH10B was used for gene cloning, *E. coli* BL21 (DE3) was used for protein expression and baicalein and scutellarein production. pET28a, pEDDuet-1 and pCDF-Duet-1 vectors were used for assembling of genes in metabolic pathway.

Baicalein and scutellarein standards were purchased from Shanghai Yuanye Biotechnology Co., Ltd. Other reagents are analytical grade reagent or chromatographic grade reagent, purchased from Sinopharm Chemical Reagent Co., Ltd.

PCR was conducted on Arktik Thermal Cycler (Thermo Fisher Scientific); ZXGP-A2050 Incubator and ZWY-211G Constant Temperature Oscillator were used for culture; high-speed freezing Centrifuge 5418R and Centrifuge 5418 (Eppendorf) were used for centrifugation. Vacuum concentration was performed with Concentrator Plus (Eppendorf); OD₆₀₀ was detected using UV-1200 Ultraviolet/Visible Spectrophotometer (Shanghai Mapada Instrument Co., Ltd.). Rotary evaporation system consists of IKA RV 10 Digital Rotary Evaporator (IKA), MZ 2C NT Chemical Diaphragm Pump and CVC3000 vacuum controller (Vacubrand). Dionex UltiMate 3000 Liquid Chromatography System (Thermo Fisher Scientific) was used for HPLC.

Liquid phase detection conditions: A phase: 0.1% formic acid solution, B phase: acetonitrile; separation conditions: 0-20 min, 20% B phase-55% B phase, 20-22 min, 55% B phase-100% B phase, 22-27 min, 100% B phase-20% B phase, 27-35 min, 100% B phase-20% B phase, 35-40 min, 20% B phase; detection wavelength: 340 nm, column temperature: 30° C. The chromatographic column was Thermo synchronis C18 RP column (250 mm*4.6 mm, 5 μ m).

Example 1. Polypeptide and its Sequence Optimization

1. Optimization of F6H Polypeptide Sequence

The sequence of *Scutellaria baicalensis* F6H (SbF6H, 517aa, Genbank access No. ASW21050.1) is:

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MELSSVIYGAIALLSLFYCYLHFSKPKKSSLNAPPEAGGARFITGHLHLM
DGRSASDKLPHINLGLLADQHGPIFTIRLGVHRAVVSSWELAKEIFTTH
DTAVMARPLIADDYLSYDGASLGFSPYGPYWEIRKLVTTTELLSARRIE
LQRATRVREITQFTGELYKLWEEKIDGSGRVLVDMKQWLGNLSNLVSRM
VVGKRFYGGDDSETTKRWGVMREFFQLIGQFIPGDGLPFLRWLDLGGFE
KRTRDTAYELDKI IAMWLAERYKREYSGDDKEQCFCMALMLS LVQANPTLQ
LHYDADTI IKATCQVLI SAASD TTTVILIWI VLSLLNADVLKVKQEELD
EQVGRERRVEESDISNLPYLQAVVKETMRLYPAPFAGVRAFSEDC TVGG
YHIQKGTFLIVNLWKLHRDPRVWSDDALEFKPQRFDDKKVEVKQDFELM

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9

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PFGGRRMCPGSLNLMHMHVFLANILQAFDITTGSTVDMTESVGLTNMK
ATPLDAILTPRLSPTLY*

Modification 1: the modified F6H mutant trF6H was constituted by removing the amino acids 2-25 of SEQ ID NO: 1 and adding two amino acids MA to the N-terminal. The sequence of trF6H is as follows (SEQ ID NO: 2):

MAMPKSSLNAPPEAGGARFITGHLHLMGRSASDKLPHINLGLLADQHG
PIFTIRLGVHRAVVSSWELAKEIFTTHDTAVMARPRLIADDYLSYDGAS
LGFSYPGYWREIRKLVTTTELLSARRIELQRATRVREITQFTGELYKLWE
EKKDGSGRVLVDMKQWLGNLNLVSRMVVGKRFYGGDDSETTKRWGRVM
REFFQLIGQFIPGDGLPFLRWLDLGGFEKTRDTRDAYELDKIIAMWLAEYR
KREYSGDDKEQCFMALMLSLVQANPTLQLHYDADTIKATCQVLI SAASD
TTTIVILWVISLNNADVLKKVQEELDEQVGRERRVEESDISNLPYLQA
VVKETMRLYPAPFAGVRAFSEDCVTGGYHIQKGTFLIVNLWKLHRDPRV
WSDDALEFKPQRFDDKKVEVKQDFELMPFGGRRMCPGSLNLMHMHVFL
ANILQAFDITTGSTVDMTESVGLTNMKATPLDAILTPRLSPTLY*

Modification 2: the modified F6H mutant 8RPtrF6H was constituted by removing the amino acids 2-25 of SEQ ID NO: 1 and adding amino acids of 8RP to the N-terminal. The sequence of 8RPtrF6H is as follows (SEQ ID NO: 3):

MALLLAVFMPKSSLNAPPEAGGARFITGHLHLMGRSASDKLPHINLGL
LADQHGPIFTIRLGVHRAVVSSWELAKEIFTTHDTAVMARPRLIADDYL
SYDGASLGFSYPGYWREIRKLVTTTELLSARRIELQRATRVREITQFTGE
LYKLWEEKDGSGRVLVDMKQWLGNLNLVSRMVVGKRFYGGDDSETTK
RWRGVMREFFQLIGQFIPGDGLPFLRWLDLGGFEKTRDTRDAYELDKIIAM
WLAEYRKREYSGDDKEQCFMALMLSLVQANPTLQLHYDADTIKATCQVL
ISAASDTTTTIVILWVISLNNADVLKKVQEELDEQVGRERRVEESDISN
LPYLQAVVKETMRLYPAPFAGVRAFSEDCVTGGYHIQKGTFLIVNLWKL
HRDPRVWSDDALEFKPQRFDDKKVEVKQDFELMPFGGRRMCPGSLNLM
HMHVFLANILQAFDITTGSTVDMTESVGLTNMKATPLDAILTPRLSPTL
Y*

Modification 3: the modified F6H mutant SumotrF6H was constituted by removing the amino acids 2-25 of SEQ ID NO: 1 and adding amino acids of Sumo to the N-terminal. The sequence of SumotrF6H is as follows (SEQ ID NO: 4):

MADSEVNQEAKPEVKPEVKPETHINLKVSDGSSEIFFKIKKTTPLRLME
AFAKRQKGEMDSLRFYDGIQADQTPEDLDMEDNDIEAHREQIGGMP
KKSSLNAPPEAGGARFITGHLHLMGRSASDKLPHINLGLLADQHGPIFT
IRLGVHRAVVSSWELAKEIFTTHDTAVMARPRLIADDYLSYDGASLGFS
PYGPYWREIRKLVTTTELLSARRIELQRATRVREITQFTGELYKLWEEKD
GSGRVLVDMKQWLGNLNLVSRMVVGKRFYGGDDSETTKRWGRVMEFF

10

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QLIGQFIPGDGLPFLRWLDLGGFEKTRDTRDAYELDKIIAMWLAEYRKREY
SGDDKEQCFMALMLSLVQANPTLQLHYDADTIKATCQVLI SAASDTTTTV
5 ILIWIISLNNADVLKKVQEELDEQVGRERRVEESDISNLPYLQAVVKE
TMRLYPAPFAGVRAFSEDCVTGGYHIQKGTFLIVNLWKLHRDPRVWSD
ALEFKPQRFDDKKVEVKQDFELMPFGGRRMCPGSLNLMHMHVFLANI
10 LQAFDITTGSTVDMTESVGLTNMKATPLDAILTPRLSPTLY*

Modification 4: the modified F6H mutant MBPtrF6H was constituted by removing the amino acids 2-25 of SEQ ID NO: 1 and adding amino acids of MBP to the N-terminal. The sequence of MBPtrF6H is as follows (SEQ ID NO: 5):

MAKIEEGKLVINGDKGYNGLAEVGKKFEKDTGIKVTVHEPDKLEEKFP
QVAATGDGPDIIFWAHDRFGGYAQSGLLAEITPDKAFQDKLYPFTWDAVR
YNGKLIAYPIAVEALSLIYNKDLLPNPKTWEEIPALDKELKAKGKSALM
FNLQEPYFTWPLIADAGGYAFKYENGKYDIKDVGVDNAGAKAGTLFLVDL
IKNKHMNADTDYSIAEAFNKGETAMTINGPWAWSNIDTSKVN YGVTVLP
25 TFKGQPSKPFVGVLSAGINAASPNKELAKEFLENYLLTDEGLEAVNKDKP
LGAVALKSYEEELVKDPRIATMENAQKGEIMPNI PQMSAFWYAVRTAVI
NAASGRQTVDEALKDAQTMPKSSLNAPPEAGGARFITGHLHLMGRSAS
DKLPHINLGLLADQHGPIFTIRLGVHRAVVSSWELAKEIFTTHDTAVMA
RPRLIADDYLSYDGASLGFSYPGYVREIRKLVTTTELLSARRIELQRAT
RVREITQFTGELYKLWEEKDGSGRVLVDMKQWLGNLNLVSRMVVGKR
35 FYGGDDSETTKRWGRVMEFFQLIGQFIPGDGLPFLRWLDLGGFEKTRD
TAYELDKIIAMWLAEYRKREYSGDDKEQCFMALMLSLVQANPTLQLHYDA
DTIIKATCQVLI SAASDTTTTIVILWVISLNNADVLKKVQEELDEQVGR
ERRVEESDISNLPYLQAVVKETMRLYPAPFAGVRAFSEDCVTGGYHIQK
40 GTFLIVNLWKLHRDPRVWSDDALEFKPQRFDDKKVEVKQDFELMPFGGG
RRMCPGSLNLMHMHVFLANILQAFDITTGSTVDMTESVGLTNMKATPLD
45 AILTPRLSPTLY*

Modification 5: the modified F6H mutant 2B1trF6H was constituted by removing the amino acids 2-25 of SEQ ID NO: 1 and adding amino acids of 2B1 to the N-terminal. The sequence of 2B1trF6H is as follows (SEQ ID NO: 6):

MAKKTSSKGKLPSPGSMPPKSSLNAPPEAGGARFITGHLHLMGRSASDK
LPHINLGLLADQHGPIFTIRLGVHRAVVSSWELAKEIFTTHDTAVMARP
55 RLIADDYLSYDGASLGFSYPGYVREIRKLVTTTELLSARRIELQRATRV
REITQFTGELYKLWEEKDGSGRVLVDMKQWLGNLNLVSRMVVGKRFY
GGDDSETTKRWGRVMEFFQLIGQFIPGDGLPFLRWLDLGGFEKTRDTR
60 YELDKIIAMWLAEYRKREYSGDDKEQCFMALMLSLVQANPTLQLHYDADT
IIKATCQVLI SAASDTTTTIVILWVISLNNADVLKKVQEELDEQVGRER
RVEESDISNLPYLQAVVKETMRLYPAPFAGVRAFSEDCVTGGYHIQKGT
65 FLIVNLWKLHRDPRVWSDDALEFKPQRFDDKKVEVKQDFELMPFGGRR

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MCPGSLNLMHMHVFLANILQAFDITTGSTVDMTESVGLTNMKATPLDAI
LTPRLSPPLY*

2. Modification of CPR

The sequence of *Arabidopsis thaliana* CPR (AtCPR, 712aa, Genebank access No. NP_849472.2) is as follows (SEQ ID NO: 7):

MSSSSSSSTSMIDLMAAIIKGPVIVSDPANASAYESVAELSSMLIENR
QFAMIVTTTIAVLIGICIVMLVWRRSGSGNSKRVEPLKPLVIKPREEIDD
GRKKVTIFFGTQTGTAEFGAKALGEEAKARYEKTRFKIVDLDYAADDE
YEEKLKKEDVAFFFLATYGDGEPTDNAARFYKWFTEGNDRGEWLKNLKYG
VFGLGNRQYEHFNKAVKVDDILVEQGAQRLVQVGLGDDQCIEDDFTAW
REALWPELDTILREEGDTAVATPYTAAVLEYRVSIHDSERAKFNDINMAN
GNGYTVFDAQHPYKANVAVKRELHTPESDRSCIHLEFDIAGSGLTYETGD
HVGVLCDNLSETVDEALRLDMSPTDYFSLHAEKEDGTPISSSLPPFPFP
CNLRTALTRYACLLSSPKKSALVALAAHASDPTAEERLKLHSLPAGKVDE
YSKWVVEQSRLLEVMFAEFPSAKPLGVFFAGVAPRLQPRFYSISSSPKI
AETRIHVTALVYEKMPGTGRIHKGVCSTWMKNVAPYEKSENCSSAPIFVR
QSNFKLPDSKVPIIMIGPGTGLAPFRGFLQERLALVESGVELGPSVLFF
GCRNRRMDFIYEELQRFVESGALAELSVAFSREGPTKEYVQHKMMDKAS
DIWNMISQGAYLYVCGDAKGMARDVHRSLSHTIAQEQGSDSTKAEGFVKV
LQTSGRYLRDVW*

The modified AtCPR mutant trAtCPR was constituted by removing the amino acids 2-72 of SEQ ID NO: 7. The sequence of trAtCPR is as follows (SEQ ID NO: 8):

MRRSGSGNSKRVEPLKPLVIKPREEIDDGRKKVTIFFGTQTGTAEFGAK
ALGEEAKARYEKTRFKIVDLDYAADDEYEEKLKKEDVAFFFLATYGDG
EPTDNAARFYKWFTEGNDRGEWLKNLKYGVFGLGNRQYEHFNKAVKVDD
ILVEQGAQRLVQVGLGDDQCIEDDFTAWREALWPELDTILREEGDTAVA
TPYTAAVLEYRVSIHDSERAKFNDINMANGNGYTVFDAQHPYKANVAVKR
ELHTPESDRSCIHLEFDIAGSGLTYETGDHVGVLCDNLSETVDEALRLD
MSPDYFSLHAEKEDGTPISSSLPPFPFPNCLRTALTRYACLLSSPKKSA
LVALAAHASDPTAEERLKLHSLPAGKVDEYSKWVVEQSRLLEVMFAEFPS
AKPPLGVFFAGVAPRLQPRFYSISSSPKIAETRIHVTALVYEKMPGTGRI
HKGVCSTWMKNVAPYEKSENCSSAPIFVRQSNFKLPDSKVPIIMIGPGT
GLAPFRGFLQERLALVESGVELGPSVLFFGCRNRRMDFIYEELQRFVES
GALAELSVAFSREGPTKEYVQHKMMDKASDIWNMISQGAYLYVCGDAKGM
ARDVHRSLSHTIAQEQGSDSTKAEGFVKVNLQTSGRYLRDVW*

Example 2. Construction of Recombinant Plasmid Containing Novel F6H Mutant

Based on pETDuet-1, plasmid pYH45 was constituted by linking AtCPR into NdeI and XhoI sites by one-step cloning method.

Based on pETDuet-1, plasmid pYH46 was constituted by linking trAtCPR into NdeI and XhoI sites by one-step cloning method.

Furthermore, pUC19-F6H was constituted by linking the codon optimized coding sequence of F6H (synthesized by GenScript) into pUC19. PCR was conducted using F6H-F/R as primers and pUC19-F6H as templates. The PCR system was 50 μ L (Primestar Max Premix, 25 μ L; final concentration 0.2-0.3 μ M of the two primers; pUC19-F6H, 0.2 μ L; the remaining volume was supplemented with sterilized distilled water). PCR reaction procedure is: pre-denaturation at 98° C. for 2 min, denaturation at 98° C. for 10 s, annealing at 55° C. for 15 s, extension at 72° C. for 20 s, 25 cycles. The amplified fragment of about 1.5 kb was detected by agarose electrophoresis, purified and digested with Nco I and BamH I. The digested fragment was ligated into pYH46 digested by the same enzymes, and the ligated product was transformed into competent cells of *E. coli* DH10B. The plasmid was extracted. The recombinant plasmid pYH59 was verified by double digestion (on restriction sites introduced during plasmid construction) and gene sequencing. Similarly, the digested fragment was ligated into pYH45 to obtain the recombinant plasmid pYH59.

PCR was conducted using trF6H-F/F6H-R as primers and pUC19-trF6H as templates. Plasmid pYH58 was obtained by ligating the amplified fragment into NdeI and BamH I of pYH46 by one-step cloning method.

PCR was conducted using 8RP-trF6H-F/F6H-R as primers and pUC19-trF6H as templates. Plasmid pYH60 was obtained by ligating the amplified fragment into NdeI and BamH I of pYH46 by one-step cloning method.

DNA fragment containing Sumo sequence was amplified using pETSumo (Invitrogen) as templates and Sumo-F/Sumo-trF6H-R as primers. DNA fragment containing trF6H was amplified using pUC19-trF6H as templates and Sumo-trF6H-F/F6H-R as primers. PCR amplification was conducted using Sumo-F/F6H-R as primers and the above two DNA fragments as templates. Plasmid pYH61 was obtained by ligating the amplified fragment into NdeI and BamH I of pYH46 by one-step cloning method.

DNA fragments containing MBP sequence was amplified using pMAL-c5x (Invitrogen) as templates and MBP-F/MBP-trF6H-R as primers. DNA fragments containing trF6H was amplified using pUC19-trF6H as templates and MBP-trF6H-F/F6H-R as primers. Then PCR amplification was conducted using MBP-F/F6H-R as primers and the above two DNA fragments as templates. Plasmid pYH62 was obtained by ligating the amplified fragment into NdeI and BamH I of pYH46 by one-step cloning method.

PCR was conducted using 2B1-F/F6H-R as primers and pUC19-trF6H as templates. Plasmid pYH63 was obtained by ligating the amplified fragment into NdeI and BamH I of pYH46 by one-step cloning method.

PCR was conducted using trF6H-F/F6H-R as primers and pUC19-trF6H as templates. Plasmid pYH64 was obtained by ligating the amplified fragment into NdeI and BamH I of pYH45 by one-step cloning method.

DNA fragments containing MBP sequence was amplified using pMAL-c5x as templates and MBP-F/MBP-trF6H-R as primers. DNA fragments containing trF6H was amplified using pUC19-trF6H as templates and MBP-trF6H-F/F6H-R as primers. Then PCR amplification was conducted using MBP-F/F6H-R as primers and the above two DNA fragments as templates. Plasmid pYH65 was obtained by ligating the amplified fragment into NdeI and BamH I of pYH45 by one-step cloning method.

PCR was conducted using 2B1-F/F6H-R as primers and pUC19-2B1trF6H as templates. Plasmid pYH66 was obtained by ligating the amplified fragment into NdeI and BamH I of pYH45 by one-step cloning method. Schematic of plasmid pYH66 is shown in FIG. 3.

The primers used in the above constructions are shown in Table 1. Schematic of the key elements in the constructed plasmid is shown in FIG. 9A.

TABLE 1

Primers	Sequences
F6H-F	TATACCATGGAACTGAGCAGTGTGA (SEQ ID NO: 9)
F6H-R	CTCGAATTCGATCCACTAGTTTAATATAAAGTCGG (SEQ ID NO: 10)
trF6H-F	CTTTAAGAAGGAGATATACCATGGCGATGCCGAAGAAAAGCTC (SEQ ID NO: 11)
8RP-trF6H-F	CTTTAAGAAGGAGATATACCATGGCTCTGTTATTAGCAGTTTTTATGCCGAAGAAAAGCTCTT (SEQ ID NO: 12)
MBP-F	CTTTAAGAAGGAGATATACCATGGCTAAAATCGAAGAAG (SEQ ID NO: 13)
MBP-trF6H-F	CTGAAAGACGCGCAGACTATGCCGAAGAAAAGCTC (SEQ ID NO: 14)
MBP-trF6H-R	GAGCTTTTCTTCGGCATAGTCTGCGCTCTTTCAG (SEQ ID NO: 15)
2B1-F	CTTTAAGAAGGAGATATACCATGGCTAAGAAAACGAGCTCTAAA GGGAAGCTCCACCAGGACCTAGCATGCCGAAGAAAAGCTCTT (SEQ ID NO: 16)
Sumo-F	CTTTAAGAAGGAGATATACCATGGCGGACTCAGAAGTCAATCTT (SEQ ID NO: 17)
Sumo-trF6H-F	GAGAACAGATTGGTGGTATGCCGAAGAAAAGCTCTT (SEQ ID NO: 18)
Sumo-trF6H-R	AAGAGCTTTTCTTCGGCATACCACCAATCTGTTCTC (SEQ ID NO: 19)

Example 3. Construction of Recombinant Plasmids Expressing PAL, 4CL, CHS, CHI and FNSI

Rhodotorula toruloides PAL (GenBankAccess No. AAA33883.1), *Petroselinum crispum* 4CL (GenBank Access No. KF765780.1), *Petunia X* hybrid CHS (GenBankAccess No. KF765781.1), *Medicago sativa* CHI gene (GenBankAccess No. KF765782.1), *Petroselinum crispum* FNS I gene (Swiss-ProtAccess No. Q7XZQ8.1) were synthesized by GenScript and constructed into pET28a, forming plasmids pET28-PAL, pET28-4CL, pET28-CHS, pET28a-CHI, and pET28a-FNSI, respectively.

The primers in Table 2 were synthesized. PCR amplification was conducted using pET28-4CL as templates and 4CL-F-NcoI/4CL-R-BamHI as primers. pYH40 was constructed by ligation of the amplified products with NcoI/BamHI digested pCDFDuet-1.

PCR amplification was conducted using pET28-CHS as templates and CHS-F-NdeI/CHS-R-XhoI as primers.

pYH50 was constructed by ligation of the amplified products with NdeI/XhoI digested pYH40.

PCR amplification was conducted using pET28a-CHI as templates and T7CHI-F-XhoI/CHI-R-AvrII as primers. Then pYH51 was constructed by ligation of the amplified products with pYH50.

PCR amplification was conducted using pET28-PAL as templates and T7PAL-F-BamH I/PAL-R-Hind III as primers. The amplified products were digested by BamH I and Hind III, and ligated with pYH51 digested by the same enzymes to form plasmid pYH55.

PCR amplification was conducted using pET28a-FNSI as templates and FNSI-HindIII-F/FNSI-NotI-R as primers. The amplified products were digested by Hind III and Not I, and ligated with pYH55 digested by the same enzymes to form plasmid pYH57. Schematic of plasmid pYH57 is shown in FIG. 4.

TABLE 2

Primers	Sequences
4CL-F-NcoI	TATACCATGGGTGACTGCGTTGCCCG (SEQ ID NO: 20)
4CL-R-BamHI	CGGGATCCTTACTTCGGCAGGTGCGCGCTC (SEQ ID NO: 21)
T7PAL-F-BamHI	CGGGATCCCTTATGCGACTCCTGCATTAG (SEQ ID NO: 22)
PAL-R-HindIII	GCCCAAGCTTTTATGCCAGCATCTTC (SEQ ID NO: 23)
CHS-F-NdeI	AGATATACATATGGTTACGGTGAAGAATAC (SEQ ID NO: 24)
CHS-R-XhoI	CCGCTCGAGTTAGGTAGCCACACTATGCAG (SEQ ID NO: 25)
T7CHI-F-XhoI	CCGCTCGAGCTAGAAATAATTTGTTTAAAC (SEQ ID NO: 26)
CHI-R-AvrII	GAGCCTAGGTTAGTTACCGATTTTAAAG (SEQ ID NO: 27)

TABLE 2-continued

Primers	Sequences
FNSI-HindIII-F	GAAGATGCTGGCATAAAAGCTTCGATCCCGCGAAATTA (SEQ ID NO: 28)
FNSI-NotI-R	CGACTTAAGCATTATGCGGCCGCTACGCCAGGTTTTC (SEQ ID NO: 29)

Example 4. Construction and Functional Verification of Baicalein and Scutellarein Synthesizing Strains

The biosynthesis process of baicalein and scutellarein is shown in FIG. 1.

The engineering strain BL21(DE3)-pYH57-pYH66 was obtained by co-transformation of the recombinant plasmids pYH66 and pYH57 into *E. coli* BL21 (DE3) competent cells.

The cells were cultured in LB solid medium (containing 80 µg/ml spectinomycin, 100 µg/ml ampicillin) overnight at 37° C. Single colony was transferred to a 2 mL LB liquid medium (containing 80 µg/ml spectinomycin, and 100 µg/ml ampicillin) and incubated overnight. The bacterial fluid was transferred to a new 10 mL MOPS liquid medium with antibiotics and incubated at 37° C. and 250 r/min until OD₆₀₀ reached 0.5-0.6. The culture was cooled down to 16° C. in a water bath. Then inducer IPTG was added at a final concentration of 1 mM, and different concentrations of sterilized phenylalanine or tyrosine was added. The mixture was placed at 22° C. for low temperature induction, and cultured for 48 h at 220 r/min. The BL21 (DE3) recombinant strain containing empty plasmid pETDuet-1 and pCDF-Duet-1 without foreign gene(s) was used as blank control, and the culture procedure was the same as above.

At the same time, the recombinant plasmids listed in Table 2 were transformed into *E. coli* and cultured to detect the production of their products.

After culture, the expression of compounds in each recombinant strain harboring recombinant plasmid was detected, which are shown in Table 3.

TABLE 3

Plasmids	Features	Uses
pYH40	expressing 4CL protein	Synthesis of pinocembrin and naringenin
pYH50	expressing 4CL and CHS proteins	
pYH51	expressing 4CL, CHS and CHI proteins	
pYH55	expressing PAL, 4CL, CHS and CHI proteins	
pYH57	expressing PAL, 4CL, CHS, CHI and FNSI proteins	Synthesis of chrysin from phenylalanine Synthesis of apigenin from tyrosine
pYH58	expressing trF6H and trCPR proteins	Synthesis of baicalein from chrysin
pYH59	expressing F6H and CPR proteins	
pYH60	expressing 8RPF6H and trCPR proteins	Synthesis of scutellarein from apigenin
pYH61	expressing SumotrF6H and trCPR proteins	
pYH62	expressing MBPtrF6H and trCPR proteins	
pYH63	expressing 2B1trF6H and trCPR proteins	
pYH64	expressing trF6H and CPR proteins	
pYH65	expressing MBPtrF6H and CPR proteins	
pYH66	expressing 2B1trF6H and CPR proteins	

It was verified that each recombinant strain of the disclosure can successfully synthesize the target compound.

HPLC results of engineering strain BL21(DE3)-pYH57-pYH66 and baicalein standard are shown in FIG. 5. Mass

spectrum results of baicalein produced by engineering strain BL21(DE3)-pYH57-pYH66 are shown in FIG. 6.

HPLC results of engineering strain BL21(DE3)-pYH57-pYH66 and scutellarein standard are shown in FIG. 7. Mass spectrum results of scutellarein produced by engineering strain BL21(DE3)-pYH57-pYH66 are shown in FIG. 8.

Example 5: Production with Chrysin as Substrate

Six recombinant plasmids (pYH58 to pYH66) were transformed into competent cells of *E. coli* BL21 (DE3) to obtain the engineering strains BL21(DE3)-pYH58 to BL21(DE3)-pYH66, respectively.

The cells were cultured in LB solid medium (containing 100 µg/ml ampicillin) overnight at 37° C. Single colony was transferred to a 2 mL LB liquid medium (containing 100 µg/ml ampicillin) and incubated overnight. The bacterial fluid was transferred to a new 20 mL MOPS liquid medium with antibiotics and incubated at 37° C. and 250 r/min until OD₆₀₀ reached 0.5-0.6. The culture was cooled down to 16° C. in a water bath. Then inducer IPTG was added at a final concentration of 1 mM. The mixture was cultured for 12 h at 22° C. and 220 r/min. After centrifugation at 6000 rpm, 4° C. for 10 min, the supernatant was removed, and the bacteria were collected and re-suspended in a reaction buffer (50 mM Tris-HCl, pH 7.4, 0.1% Triton) until OD₆₀₀ reached 30.5 µL chrysin (25 mM) and 2.5 µL NADPH (100 mM) were added to 1 mL of the suspension of the recombinant bacteria, and the reaction was continued at 37° C. for 8 hours. After completion of the reaction, the solution was extracted for 3 times by 10 µL HCl (6 M) and 1 mL ethyl acetate. The

organic phase was concentrated, and the resulting residue was dissolved with 200 µL methanol, wherein 10 µL was used for HPLC analysis.

The conversion rates of baicalein from chrysin in each recombinant *E. coli* were shown in FIG. 9B.

HPLC results of the catalytic reaction solution of each recombinant *E. coli* were shown in FIG. 9C.

Each reference provided herein is incorporated by reference to the same extent as if each reference was individually incorporated by reference. In addition, it should be under-

stood that based on the above teaching content of the disclosure, those skilled in the art can practice various changes or modifications to the disclosure, and these equivalent forms also fall within the scope of the appended claims.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 29

<210> SEQ ID NO 1

<211> LENGTH: 517

<212> TYPE: PRT

<213> ORGANISM: *Scutellaria baicalensis*

<400> SEQUENCE: 1

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20          25          30

Ala Pro Pro Glu Ala Gly Gly Ala Arg Phe Ile Thr Gly His Leu His
35          40          45

Leu Met Asp Gly Arg Ser Ala Ser Asp Lys Leu Pro His Ile Asn Leu
50          55          60

Gly Leu Leu Ala Asp Gln His Gly Pro Ile Phe Thr Ile Arg Leu Gly
65          70          75          80

Val His Arg Ala Val Val Val Ser Ser Trp Glu Leu Ala Lys Glu Ile
85          90          95

Phe Thr Thr His Asp Thr Ala Val Met Ala Arg Pro Arg Leu Ile Ala
100         105         110

Asp Asp Tyr Leu Ser Tyr Asp Gly Ala Ser Leu Gly Phe Ser Pro Tyr
115         120         125

Gly Pro Tyr Trp Arg Glu Ile Arg Lys Leu Val Thr Thr Glu Leu Leu
130         135         140

Ser Ala Arg Arg Ile Glu Leu Gln Arg Ala Thr Arg Val Arg Glu Ile
145         150         155         160

Thr Gln Phe Thr Gly Glu Leu Tyr Lys Leu Trp Glu Glu Lys Lys Asp
165         170         175

Gly Ser Gly Arg Val Leu Val Asp Met Lys Gln Trp Leu Gly Asn Leu
180         185         190

Ser Leu Asn Leu Val Ser Arg Met Val Val Gly Lys Arg Phe Tyr Gly
195         200         205

Gly Asp Asp Ser Glu Thr Thr Lys Arg Trp Arg Gly Val Met Arg Glu
210         215         220

Phe Phe Gln Leu Ile Gly Gln Phe Ile Pro Gly Asp Gly Leu Pro Phe
225         230         235         240

Leu Arg Trp Leu Asp Leu Gly Gly Phe Glu Lys Arg Thr Arg Asp Thr
245         250         255

Ala Tyr Glu Leu Asp Lys Ile Ile Ala Met Trp Leu Ala Glu Tyr Arg
260         265         270

Lys Arg Glu Tyr Ser Gly Asp Asp Lys Glu Gln Cys Phe Met Ala Leu
275         280         285

Met Leu Ser Leu Val Gln Ala Asn Pro Thr Leu Gln Leu His Tyr Asp
290         295         300

Ala Asp Thr Ile Ile Lys Ala Thr Cys Gln Val Leu Ile Ser Ala Ala
305         310         315         320

Ser Asp Thr Thr Thr Val Ile Leu Ile Trp Val Ile Ser Leu Leu Leu
325         330         335

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Asn Asn Ala Asp Val Leu Lys Lys Val Gln Glu Glu Leu Asp Glu Gln
   340                               345                   350

Val Gly Arg Glu Arg Arg Val Glu Glu Ser Asp Ile Ser Asn Leu Pro
   355                               360                   365

Tyr Leu Gln Ala Val Val Lys Glu Thr Met Arg Leu Tyr Pro Pro Ala
   370                               375                   380

Pro Phe Ala Gly Val Arg Ala Phe Ser Glu Asp Cys Thr Val Gly Gly
   385                               390                   395                   400

Tyr His Ile Gln Lys Gly Thr Phe Leu Ile Val Asn Leu Trp Lys Leu
   405                               410                   415

His Arg Asp Pro Arg Val Trp Ser Asp Asp Ala Leu Glu Phe Lys Pro
   420                               425                   430

Gln Arg Phe Phe Asp Lys Lys Val Glu Val Lys Gly Gln Asp Phe Glu
   435                               440                   445

Leu Met Pro Phe Gly Gly Gly Arg Arg Met Cys Pro Gly Ser Asn Leu
   450                               455                   460

Gly Met His Met Val His Phe Val Leu Ala Asn Ile Leu Gln Ala Phe
   465                               470                   475                   480

Asp Ile Thr Thr Gly Ser Thr Val Asp Met Thr Glu Ser Val Gly Leu
   485                               490                   495

Thr Asn Met Lys Ala Thr Pro Leu Asp Ala Ile Leu Thr Pro Arg Leu
   500                               505                   510

Ser Pro Thr Leu Tyr
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<210> SEQ ID NO 2
<211> LENGTH: 495
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: F6H mutant

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<400> SEQUENCE: 2

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Met Ala Met Pro Lys Lys Ser Ser Leu Asn Ala Pro Pro Glu Ala Gly
 1           5           10           15

Gly Ala Arg Phe Ile Thr Gly His Leu His Leu Met Asp Gly Arg Ser
 20           25           30

Ala Ser Asp Lys Leu Pro His Ile Asn Leu Gly Leu Leu Ala Asp Gln
 35           40           45

His Gly Pro Ile Phe Thr Ile Arg Leu Gly Val His Arg Ala Val Val
 50           55           60

Val Ser Ser Trp Glu Leu Ala Lys Glu Ile Phe Thr Thr His Asp Thr
 65           70           75           80

Ala Val Met Ala Arg Pro Arg Leu Ile Ala Asp Asp Tyr Leu Ser Tyr
 85           90           95

Asp Gly Ala Ser Leu Gly Phe Ser Pro Tyr Gly Pro Tyr Trp Arg Glu
100           105           110

Ile Arg Lys Leu Val Thr Thr Glu Leu Leu Ser Ala Arg Arg Ile Glu
115           120           125

Leu Gln Arg Ala Thr Arg Val Arg Glu Ile Thr Gln Phe Thr Gly Glu
130           135           140

Leu Tyr Lys Leu Trp Glu Glu Lys Lys Asp Gly Ser Gly Arg Val Leu
145           150           155           160

Val Asp Met Lys Gln Trp Leu Gly Asn Leu Ser Leu Asn Leu Val Ser
165           170           175

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Arg Met Val Val Gly Lys Arg Phe Tyr Gly Gly Asp Asp Ser Glu Thr
      180                      185                      190

Thr Lys Arg Trp Arg Gly Val Met Arg Glu Phe Phe Gln Leu Ile Gly
      195                      200                      205

Gln Phe Ile Pro Gly Asp Gly Leu Pro Phe Leu Arg Trp Leu Asp Leu
      210                      215                      220

Gly Gly Phe Glu Lys Arg Thr Arg Asp Thr Ala Tyr Glu Leu Asp Lys
      225                      230                      235                      240

Ile Ile Ala Met Trp Leu Ala Glu Tyr Arg Lys Arg Glu Tyr Ser Gly
      245                      250                      255

Asp Asp Lys Glu Gln Cys Phe Met Ala Leu Met Leu Ser Leu Val Gln
      260                      265                      270

Ala Asn Pro Thr Leu Gln Leu His Tyr Asp Ala Asp Thr Ile Ile Lys
      275                      280                      285

Ala Thr Cys Gln Val Leu Ile Ser Ala Ala Ser Asp Thr Thr Thr Val
      290                      295                      300

Ile Leu Ile Trp Val Ile Ser Leu Leu Leu Asn Asn Ala Asp Val Leu
      305                      310                      315                      320

Lys Lys Val Gln Glu Glu Leu Asp Glu Gln Val Gly Arg Glu Arg Arg
      325                      330                      335

Val Glu Glu Ser Asp Ile Ser Asn Leu Pro Tyr Leu Gln Ala Val Val
      340                      345                      350

Lys Glu Thr Met Arg Leu Tyr Pro Pro Ala Pro Phe Ala Gly Val Arg
      355                      360                      365

Ala Phe Ser Glu Asp Cys Thr Val Gly Gly Tyr His Ile Gln Lys Gly
      370                      375                      380

Thr Phe Leu Ile Val Asn Leu Trp Lys Leu His Arg Asp Pro Arg Val
      385                      390                      395                      400

Trp Ser Asp Asp Ala Leu Glu Phe Lys Pro Gln Arg Phe Phe Asp Lys
      405                      410                      415

Lys Val Glu Val Lys Gly Gln Asp Phe Glu Leu Met Pro Phe Gly Gly
      420                      425                      430

Gly Arg Arg Met Cys Pro Gly Ser Asn Leu Gly Met His Met Val His
      435                      440                      445

Phe Val Leu Ala Asn Ile Leu Gln Ala Phe Asp Ile Thr Thr Gly Ser
      450                      455                      460

Thr Val Asp Met Thr Glu Ser Val Gly Leu Thr Asn Met Lys Ala Thr
      465                      470                      475                      480

Pro Leu Asp Ala Ile Leu Thr Pro Arg Leu Ser Pro Thr Leu Tyr
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<210> SEQ ID NO 3
<211> LENGTH: 501
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: F6H mutant 8RPTrF6H

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<400> SEQUENCE: 3

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Ala Pro Pro Glu Ala Gly Gly Ala Arg Phe Ile Thr Gly His Leu His
20          25          30

Leu Met Asp Gly Arg Ser Ala Ser Asp Lys Leu Pro His Ile Asn Leu
35          40          45

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Gly 50	Leu	Ala	Asp	Gln	His	Gly	Pro	Ile	Phe	Thr	Ile	Arg	Leu	Gly
Val 65	His	Arg	Ala	Val	Val	Val	Ser	Ser	Trp	Glu	Leu	Ala	Lys	Ile
Phe	Thr	Thr	His	Asp	Thr	Ala	Val	Met	Ala	Arg	Pro	Arg	Leu	Ala
Asp	Asp	Tyr	Leu	Ser	Tyr	Asp	Gly	Ala	Ser	Leu	Gly	Phe	Ser	Tyr
Gly	Pro	Tyr	Trp	Arg	Glu	Ile	Arg	Lys	Leu	Val	Thr	Thr	Glu	Leu
Ser	Ala	Arg	Arg	Ile	Glu	Leu	Gln	Arg	Ala	Thr	Arg	Val	Arg	Ile
Thr	Gln	Phe	Thr	Gly	Glu	Leu	Tyr	Lys	Leu	Trp	Glu	Glu	Lys	Asp
Gly	Ser	Gly	Arg	Val	Leu	Val	Asp	Met	Lys	Gln	Trp	Leu	Gly	Leu
Ser	Leu	Asn	Leu	Val	Ser	Arg	Met	Val	Val	Gly	Lys	Arg	Phe	Gly
Gly	Asp	Asp	Ser	Glu	Thr	Thr	Lys	Arg	Trp	Arg	Gly	Val	Met	Glu
Phe	Phe	Gln	Leu	Ile	Gly	Gln	Phe	Ile	Pro	Gly	Asp	Gly	Leu	Phe
Leu	Arg	Trp	Leu	Asp	Leu	Gly	Gly	Phe	Glu	Lys	Arg	Thr	Arg	Thr
Ala	Tyr	Glu	Leu	Asp	Lys	Ile	Ile	Ala	Met	Trp	Leu	Ala	Glu	Arg
Lys	Arg	Glu	Tyr	Ser	Gly	Asp	Asp	Lys	Glu	Gln	Cys	Phe	Met	Leu
Met	Leu	Ser	Leu	Val	Gln	Ala	Asn	Pro	Thr	Leu	Gln	Leu	His	Asp
Ala	Asp	Thr	Ile	Ile	Lys	Ala	Thr	Cys	Gln	Val	Leu	Ile	Ser	Ala
Ser	Asp	Thr	Thr	Thr	Val	Ile	Leu	Ile	Trp	Val	Ile	Ser	Leu	Leu
Asn	Asn	Ala	Asp	Val	Leu	Lys	Lys	Val	Gln	Glu	Glu	Leu	Asp	Gln
Val	Gly	Arg	Glu	Arg	Arg	Val	Glu	Glu	Ser	Asp	Ile	Ser	Asn	Pro
Tyr	Leu	Gln	Ala	Val	Val	Lys	Glu	Thr	Met	Arg	Leu	Tyr	Pro	Ala
Pro	Phe	Ala	Gly	Val	Arg	Ala	Phe	Ser	Glu	Asp	Cys	Thr	Val	Gly
Tyr	His	Ile	Gln	Lys	Gly	Thr	Phe	Leu	Ile	Val	Asn	Leu	Trp	Leu
His	Arg	Asp	Pro	Arg	Val	Trp	Ser	Asp	Asp	Ala	Leu	Glu	Phe	Pro
Gln	Arg	Phe	Phe	Asp	Lys	Lys	Val	Glu	Val	Lys	Gly	Gln	Asp	Glu
Leu	Met	Pro	Phe	Gly	Gly	Gly	Arg	Arg	Met	Cys	Pro	Gly	Ser	Leu
Gly	Met	His	Met	Val	His	Phe	Val	Leu	Ala	Asn	Ile	Leu	Gln	Phe

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Asp Ile Thr Thr Gly Ser Thr Val Asp Met Thr Glu Ser Val Gly Leu
 465 470 475 480

Thr Asn Met Lys Ala Thr Pro Leu Asp Ala Ile Leu Thr Pro Arg Leu
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Ser Pro Thr Leu Tyr
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<210> SEQ ID NO 4
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 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: F6H mutant SumotrF6H

<400> SEQUENCE: 4

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 20 25 30

Ser Glu Ile Phe Phe Lys Ile Lys Lys Thr Thr Pro Leu Arg Arg Leu
 35 40 45

Met Glu Ala Phe Ala Lys Arg Gln Gly Lys Glu Met Asp Ser Leu Arg
 50 55 60

Phe Leu Tyr Asp Gly Ile Arg Ile Gln Ala Asp Gln Thr Pro Glu Asp
 65 70 75 80

Leu Asp Met Glu Asp Asn Asp Ile Ile Glu Ala His Arg Glu Gln Ile
 85 90 95

Gly Gly Met Pro Lys Lys Ser Ser Leu Asn Ala Pro Pro Glu Ala Gly
 100 105 110

Gly Ala Arg Phe Ile Thr Gly His Leu His Leu Met Asp Gly Arg Ser
 115 120 125

Ala Ser Asp Lys Leu Pro His Ile Asn Leu Gly Leu Leu Ala Asp Gln
 130 135 140

His Gly Pro Ile Phe Thr Ile Arg Leu Gly Val His Arg Ala Val Val
 145 150 155 160

Val Ser Ser Trp Glu Leu Ala Lys Glu Ile Phe Thr Thr His Asp Thr
 165 170 175

Ala Val Met Ala Arg Pro Arg Leu Ile Ala Asp Asp Tyr Leu Ser Tyr
 180 185 190

Asp Gly Ala Ser Leu Gly Phe Ser Pro Tyr Gly Pro Tyr Trp Arg Glu
 195 200 205

Ile Arg Lys Leu Val Thr Thr Glu Leu Leu Ser Ala Arg Arg Ile Glu
 210 215 220

Leu Gln Arg Ala Thr Arg Val Arg Glu Ile Thr Gln Phe Thr Gly Glu
 225 230 235 240

Leu Tyr Lys Leu Trp Glu Glu Lys Lys Asp Gly Ser Gly Arg Val Leu
 245 250 255

Val Asp Met Lys Gln Trp Leu Gly Asn Leu Ser Leu Asn Leu Val Ser
 260 265 270

Arg Met Val Val Gly Lys Arg Phe Tyr Gly Gly Asp Asp Ser Glu Thr
 275 280 285

Thr Lys Arg Trp Arg Gly Val Met Arg Glu Phe Phe Gln Leu Ile Gly
 290 295 300

Gln Phe Ile Pro Gly Asp Gly Leu Pro Phe Leu Arg Trp Leu Asp Leu
 305 310 315 320

Gly	Gly	Phe	Glu	Lys	Arg	Thr	Arg	Asp	Thr	Ala	Tyr	Glu	Leu	Asp	Lys
				325					330					335	
Ile	Ile	Ala	Met	Trp	Leu	Ala	Glu	Tyr	Arg	Lys	Arg	Glu	Tyr	Ser	Gly
			340					345					350		
Asp	Asp	Lys	Glu	Gln	Cys	Phe	Met	Ala	Leu	Met	Leu	Ser	Leu	Val	Gln
		355					360					365			
Ala	Asn	Pro	Thr	Leu	Gln	Leu	His	Tyr	Asp	Ala	Asp	Thr	Ile	Ile	Lys
	370						375				380				
Ala	Thr	Cys	Gln	Val	Leu	Ile	Ser	Ala	Ala	Ser	Asp	Thr	Thr	Thr	Val
385					390					395					400
Ile	Leu	Ile	Trp	Val	Ile	Ser	Leu	Leu	Leu	Asn	Asn	Ala	Asp	Val	Leu
				405					410					415	
Lys	Lys	Val	Gln	Glu	Glu	Leu	Asp	Glu	Gln	Val	Gly	Arg	Glu	Arg	Arg
			420					425					430		
Val	Glu	Glu	Ser	Asp	Ile	Ser	Asn	Leu	Pro	Tyr	Leu	Gln	Ala	Val	Val
		435					440					445			
Lys	Glu	Thr	Met	Arg	Leu	Tyr	Pro	Pro	Ala	Pro	Phe	Ala	Gly	Val	Arg
	450					455					460				
Ala	Phe	Ser	Glu	Asp	Cys	Thr	Val	Gly	Gly	Tyr	His	Ile	Gln	Lys	Gly
465					470					475					480
Thr	Phe	Leu	Ile	Val	Asn	Leu	Trp	Lys	Leu	His	Arg	Asp	Pro	Arg	Val
				485					490					495	
Trp	Ser	Asp	Asp	Ala	Leu	Glu	Phe	Lys	Pro	Gln	Arg	Phe	Phe	Asp	Lys
			500					505					510		
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		515					520					525			
Gly	Arg	Arg	Met	Cys	Pro	Gly	Ser	Asn	Leu	Gly	Met	His	Met	Val	His
	530					535						540			
Phe	Val	Leu	Ala	Asn	Ile	Leu	Gln	Ala	Phe	Asp	Ile	Thr	Thr	Gly	Ser
545					550					555					560
Thr	Val	Asp	Met	Thr	Glu	Ser	Val	Gly	Leu	Thr	Asn	Met	Lys	Ala	Thr
				565					570					575	
Pro	Leu	Asp	Ala	Ile	Leu	Thr	Pro	Arg	Leu	Ser	Pro	Thr	Leu	Tyr	
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<211> LENGTH: 861
<212> TYPE: PRT
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<220> FEATURE:
<223> OTHER INFORMATION: F6H mutant MBPtrF6H
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			20					25					30		
Thr	Gly	Ile	Lys	Val	Thr	Val	Glu	His	Pro	Asp	Lys	Leu	Glu	Glu	Lys
		35					40					45			
Phe	Pro	Gln	Val	Ala	Ala	Thr	Gly	Asp	Gly	Pro	Asp	Ile	Ile	Phe	Trp
	50					55					60				
Ala	His	Asp	Arg	Phe	Gly	Gly	Tyr	Ala	Gln	Ser	Gly	Leu	Leu	Ala	Glu
65					70					75					80
Ile	Thr	Pro	Asp	Lys	Ala	Phe	Gln	Asp	Lys	Leu	Tyr	Pro	Phe	Thr	Trp
			85						90					95	

Asp	Ala	Val	Arg	Tyr	Asn	Gly	Lys	Leu	Ile	Ala	Tyr	Pro	Ile	Ala	Val
			100					105					110		
Glu	Ala	Leu	Ser	Leu	Ile	Tyr	Asn	Lys	Asp	Leu	Leu	Pro	Asn	Pro	Pro
		115					120					125			
Lys	Thr	Trp	Glu	Glu	Ile	Pro	Ala	Leu	Asp	Lys	Glu	Leu	Lys	Ala	Lys
		130				135					140				
Gly	Lys	Ser	Ala	Leu	Met	Phe	Asn	Leu	Gln	Glu	Pro	Tyr	Phe	Thr	Trp
				150						155					160
Pro	Leu	Ile	Ala	Ala	Asp	Gly	Gly	Tyr	Ala	Phe	Lys	Tyr	Glu	Asn	Gly
				165					170					175	
Lys	Tyr	Asp	Ile	Lys	Asp	Val	Gly	Val	Asp	Asn	Ala	Gly	Ala	Lys	Ala
			180					185					190		
Gly	Leu	Thr	Phe	Leu	Val	Asp	Leu	Ile	Lys	Asn	Lys	His	Met	Asn	Ala
		195					200					205			
Asp	Thr	Asp	Tyr	Ser	Ile	Ala	Glu	Ala	Ala	Phe	Asn	Lys	Gly	Glu	Thr
						215					220				
Ala	Met	Thr	Ile	Asn	Gly	Pro	Trp	Ala	Trp	Ser	Asn	Ile	Asp	Thr	Ser
				230						235					240
Lys	Val	Asn	Tyr	Gly	Val	Thr	Val	Leu	Pro	Thr	Phe	Lys	Gly	Gln	Pro
				245					250					255	
Ser	Lys	Pro	Phe	Val	Gly	Val	Leu	Ser	Ala	Gly	Ile	Asn	Ala	Ala	Ser
			260					265					270		
Pro	Asn	Lys	Glu	Leu	Ala	Lys	Glu	Phe	Leu	Glu	Asn	Tyr	Leu	Leu	Thr
		275					280					285			
Asp	Glu	Gly	Leu	Glu	Ala	Val	Asn	Lys	Asp	Lys	Pro	Leu	Gly	Ala	Val
		290				295					300				
Ala	Leu	Lys	Ser	Tyr	Glu	Glu	Glu	Leu	Val	Lys	Asp	Pro	Arg	Ile	Ala
					310					315					320
Ala	Thr	Met	Glu	Asn	Ala	Gln	Lys	Gly	Glu	Ile	Met	Pro	Asn	Ile	Pro
				325					330					335	
Gln	Met	Ser	Ala	Phe	Trp	Tyr	Ala	Val	Arg	Thr	Ala	Val	Ile	Asn	Ala
			340					345					350		
Ala	Ser	Gly	Arg	Gln	Thr	Val	Asp	Glu	Ala	Leu	Lys	Asp	Ala	Gln	Thr
		355					360					365			
Met	Pro	Lys	Lys	Ser	Ser	Leu	Asn	Ala	Pro	Pro	Glu	Ala	Gly	Gly	Ala
						375					380				
Arg	Phe	Ile	Thr	Gly	His	Leu	His	Leu	Met	Asp	Gly	Arg	Ser	Ala	Ser
					390					395					400
Asp	Lys	Leu	Pro	His	Ile	Asn	Leu	Gly	Leu	Leu	Ala	Asp	Gln	His	Gly
				405					410					415	
Pro	Ile	Phe	Thr	Ile	Arg	Leu	Gly	Val	His	Arg	Ala	Val	Val	Val	Ser
			420					425				430			
Ser	Trp	Glu	Leu	Ala	Lys	Glu	Ile	Phe	Thr	Thr	His	Asp	Thr	Ala	Val
		435					440					445			
Met	Ala	Arg													

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515					520					525					
Met	Lys	Gln	Trp	Leu	Gly	Asn	Leu	Ser	Leu	Asn	Leu	Val	Ser	Arg	Met
530						535					540				
Val	Val	Gly	Lys	Arg	Phe	Tyr	Gly	Gly	Asp	Asp	Ser	Glu	Thr	Thr	Lys
545					550					555					560
Arg	Trp	Arg	Gly	Val	Met	Arg	Glu	Phe	Phe	Gln	Leu	Ile	Gly	Gln	Phe
				565						570				575	
Ile	Pro	Gly	Asp	Gly	Leu	Pro	Phe	Leu	Arg	Trp	Leu	Asp	Leu	Gly	Gly
			580					585					590		
Phe	Glu	Lys	Arg	Thr	Arg	Asp	Thr	Ala	Tyr	Glu	Leu	Asp	Lys	Ile	Ile
		595					600					605			
Ala	Met	Trp	Leu	Ala	Glu	Tyr	Arg	Lys	Arg	Glu	Tyr	Ser	Gly	Asp	Asp
	610					615					620				
Lys	Glu	Gln	Cys	Phe	Met	Ala	Leu	Met	Leu	Ser	Leu	Val	Gln	Ala	Asn
625					630					635					640
Pro	Thr	Leu	Gln	Leu	His	Tyr	Asp	Ala	Asp	Thr	Ile	Ile	Lys	Ala	Thr
				645					650					655	
Cys	Gln	Val	Leu	Ile	Ser	Ala	Ala	Ser	Asp	Thr	Thr	Thr	Val	Ile	Leu
			660					665					670		
Ile	Trp	Val	Ile	Ser	Leu	Leu	Leu	Asn	Asn	Ala	Asp	Val	Leu	Lys	Lys
		675					680					685			
Val	Gln	Glu	Glu	Leu	Asp	Glu	Gln	Val	Gly	Arg	Glu	Arg	Arg	Val	Glu
	690					695					700				
Glu	Ser	Asp	Ile	Ser	Asn	Leu	Pro	Tyr	Leu	Gln	Ala	Val	Val	Lys	Glu
705					710					715					720
Thr	Met	Arg	Leu	Tyr	Pro	Pro	Ala	Pro	Phe	Ala	Gly	Val	Arg	Ala	Phe
				725					730					735	
Ser	Glu	Asp	Cys	Thr	Val	Gly	Gly	Tyr	His	Ile	Gln	Lys	Gly	Thr	Phe
			740					745					750		
Leu	Ile	Val	Asn	Leu	Trp	Lys	Leu	His	Arg	Asp	Pro	Arg	Val	Trp	Ser
		755					760					765			
Asp	Asp	Ala	Leu	Glu	Phe	Lys	Pro	Gln	Arg	Phe	Phe	Asp	Lys	Lys	Val
	770					775					780				
Glu	Val	Lys	Gly	Gln	Asp	Phe	Glu	Leu	Met	Pro	Phe	Gly	Gly	Gly	Arg
785					790					795					800
Arg	Met	Cys	Pro	Gly	Ser	Asn	Leu	Gly	Met	His	Met	Val	His	Phe	Val
				805					810					815	
Leu	Ala	Asn	Ile	Leu	Gln	Ala	Phe	Asp	Ile	Thr	Thr	Gly	Ser	Thr	Val
			820					825					830		
Asp	Met	Thr	Glu	Ser	Val	Gly	Leu	Thr	Asn	Met	Lys	Ala	Thr	Pro	Leu
	835						840					845			
Asp	Ala	Ile	Leu	Thr	Pro	Arg	Leu	Ser	Pro	Thr	Leu	Tyr			
	850					855					860				

<210> SEQ ID NO 6

<211> LENGTH: 509

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: F6H mutant 2B1trF6H

<400> SEQUENCE: 6

Met	Ala	Lys	Lys	Thr	Ser	Ser	Lys	Gly	Lys	Leu	Pro	Pro	Gly	Pro	Ser
1				5						10				15	

Met Pro Lys Lys Ser Ser Leu Asn Ala Pro Pro Glu Ala Gly Gly Ala

-continued

20						25						30					
Arg	Phe	Ile	Thr	Gly	His	Leu	His	Leu	Met	Asp	Gly	Arg	Ser	Ala	Ser		
		35					40					45					
Asp	Lys	Leu	Pro	His	Ile	Asn	Leu	Gly	Leu	Leu	Ala	Asp	Gln	His	Gly		
	50					55					60						
Pro	Ile	Phe	Thr	Ile	Arg	Leu	Gly	Val	His	Arg	Ala	Val	Val	Val	Ser		
65					70					75					80		
Ser	Trp	Glu	Leu	Ala	Lys	Glu	Ile	Phe	Thr	Thr	His	Asp	Thr	Ala	Val		
				85					90					95			
Met	Ala	Arg	Pro	Arg	Leu	Ile	Ala	Asp	Asp	Tyr	Leu	Ser	Tyr	Asp	Gly		
			100					105					110				
Ala	Ser	Leu	Gly	Phe	Ser	Pro	Tyr	Gly	Pro	Tyr	Trp	Arg	Glu	Ile	Arg		
		115					120					125					
Lys	Leu	Val	Thr	Thr	Glu	Leu	Leu	Ser	Ala	Arg	Arg	Ile	Glu	Leu	Gln		
	130					135					140						
Arg	Ala	Thr	Arg	Val	Arg	Glu	Ile	Thr	Gln	Phe	Thr	Gly	Glu	Leu	Tyr		
145					150					155					160		
Lys	Leu	Trp	Glu	Glu	Lys	Lys	Asp	Gly	Ser	Gly	Arg	Val	Leu	Val	Asp		
			165					170						175			
Met	Lys	Gln	Trp	Leu	Gly	Asn	Leu	Ser	Leu	Asn	Leu	Val	Ser	Arg	Met		
			180					185					190				
Val	Val	Gly	Lys	Arg	Phe	Tyr	Gly	Gly	Asp	Asp	Ser	Glu	Thr	Thr	Lys		
		195					200					205					
Arg	Trp	Arg	Gly	Val	Met	Arg	Glu	Phe	Phe	Gln	Leu	Ile	Gly	Gln	Phe		
	210					215					220						
Ile	Pro	Gly	Asp	Gly	Leu	Pro	Phe	Leu	Arg	Trp	Leu	Asp	Leu	Gly	Gly		
225					230					235				240			
Phe	Glu	Lys	Arg	Thr	Arg	Asp	Thr	Ala	Tyr	Glu	Leu	Asp	Lys	Ile	Ile		
			245						250					255			
Ala	Met	Trp	Leu	Ala	Glu	Tyr	Arg	Lys	Arg	Glu	Tyr	Ser	Gly	Asp	Asp		
		260						265					270				
Lys	Glu	Gln	Cys	Phe	Met	Ala	Leu	Met	Leu	Ser	Leu	Val	Gln	Ala	Asn		
		275					280					285					
Pro	Thr	Leu	Gln	Leu	His	Tyr	Asp	Ala	Asp	Thr	Ile	Ile	Lys	Ala	Thr		
	290					295					300						
Cys	Gln	Val	Leu	Ile	Ser	Ala	Ala	Ser	Asp	Thr	Thr	Thr	Val	Ile	Leu		
305					310					315					320		
Ile	Trp	Val	Ile	Ser	Leu	Leu	Leu	Asn	Asn	Ala	Asp	Val	Leu	Lys	Lys		
			325					330						335			
Val	Gln	Glu	Glu	Leu	Asp	Glu	Gln	Val	Gly	Arg	Glu	Arg	Arg	Val	Glu		
		340					345						350				
Glu	Ser	Asp	Ile	Ser	Asn	Leu	Pro	Tyr	Leu	Gln	Ala	Val	Val	Lys	Glu		
		355				360						365					
Thr	Met	Arg	Leu	Tyr	Pro	Pro	Ala	Pro	Phe	Ala	Gly	Val	Arg	Ala	Phe		
	370					375					380						
Ser	Glu	Asp	Cys	Thr	Val	Gly	Gly	Tyr	His	Ile	Gln	Lys	Gly	Thr	Phe		
385					390					395					400		
Leu	Ile	Val	Asn	Leu	Trp	Lys	Leu	His	Arg	Asp	Pro	Arg	Val	Trp	Ser		
			405						410					415			
Asp	Asp	Ala	Leu	Glu	Phe	Lys	Pro	Gln	Arg	Phe	Phe	Asp	Lys	Lys	Val		
		420						425					430				
Glu	Val	Lys	Gly	Gln	Asp	Phe	Glu	Leu	Met	Pro	Phe	Gly	Gly	Gly	Arg		
		435					440					445					

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Arg Met Cys Pro Gly Ser Asn Leu Gly Met His Met Val His Phe Val
  450                      455                      460

Leu Ala Asn Ile Leu Gln Ala Phe Asp Ile Thr Thr Gly Ser Thr Val
  465                      470                      475                      480

Asp Met Thr Glu Ser Val Gly Leu Thr Asn Met Lys Ala Thr Pro Leu
      485                      490                      495

Asp Ala Ile Leu Thr Pro Arg Leu Ser Pro Thr Leu Tyr
      500                      505

<210> SEQ ID NO 7
<211> LENGTH: 712
<212> TYPE: PRT
<213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 7

Met Ser Ser Ser Ser Ser Ser Ser Thr Ser Met Ile Asp Leu Met Ala
  1                      5                      10                      15

Ala Ile Ile Lys Gly Glu Pro Val Ile Val Ser Asp Pro Ala Asn Ala
      20                      25                      30

Ser Ala Tyr Glu Ser Val Ala Ala Glu Leu Ser Ser Met Leu Ile Glu
      35                      40                      45

Asn Arg Gln Phe Ala Met Ile Val Thr Thr Ser Ile Ala Val Leu Ile
      50                      55                      60

Gly Cys Ile Val Met Leu Val Trp Arg Arg Ser Gly Ser Gly Asn Ser
      65                      70                      75                      80

Lys Arg Val Glu Pro Leu Lys Pro Leu Val Ile Lys Pro Arg Glu Glu
      85                      90                      95

Glu Ile Asp Asp Gly Arg Lys Lys Val Thr Ile Phe Phe Gly Thr Gln
      100                     105                     110

Thr Gly Thr Ala Glu Gly Phe Ala Lys Ala Leu Gly Glu Glu Ala Lys
      115                     120                     125

Ala Arg Tyr Glu Lys Thr Arg Phe Lys Ile Val Asp Leu Asp Asp Tyr
      130                     135                     140

Ala Ala Asp Asp Asp Glu Tyr Glu Glu Lys Leu Lys Lys Glu Asp Val
      145                     150                     155                     160

Ala Phe Phe Phe Leu Ala Thr Tyr Gly Asp Gly Glu Pro Thr Asp Asn
      165                     170                     175

Ala Ala Arg Phe Tyr Lys Trp Phe Thr Glu Gly Asn Asp Arg Gly Glu
      180                     185                     190

Trp Leu Lys Asn Leu Lys Tyr Gly Val Phe Gly Leu Gly Asn Arg Gln
      195                     200                     205

Tyr Glu His Phe Asn Lys Val Ala Lys Val Val Asp Asp Ile Leu Val
      210                     215                     220

Glu Gln Gly Ala Gln Arg Leu Val Gln Val Gly Leu Gly Asp Asp Asp
      225                     230                     235                     240

Gln Cys Ile Glu Asp Asp Phe Thr Ala Trp Arg Glu Ala Leu Trp Pro
      245                     250                     255

Glu Leu Asp Thr Ile Leu Arg Glu Glu Gly Asp Thr Ala Val Ala Thr
      260                     265                     270

Pro Tyr Thr Ala Ala Val Leu Glu Tyr Arg Val Ser Ile His Asp Ser
      275                     280                     285

Glu Asp Ala Lys Phe Asn Asp Ile Asn Met Ala Asn Gly Asn Gly Tyr
      290                     295                     300

Thr Val Phe Asp Ala Gln His Pro Tyr Lys Ala Asn Val Ala Val Lys

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305	310	315	320
Arg Glu Leu His Thr Pro Glu Ser Asp Arg Ser Cys Ile His Leu Glu			
	325	330	335
Phe Asp Ile Ala Gly Ser Gly Leu Thr Tyr Glu Thr Gly Asp His Val			
	340	345	350
Gly Val Leu Cys Asp Asn Leu Ser Glu Thr Val Asp Glu Ala Leu Arg			
	355	360	365
Leu Leu Asp Met Ser Pro Asp Thr Tyr Phe Ser Leu His Ala Glu Lys			
	370	375	380
Glu Asp Gly Thr Pro Ile Ser Ser Ser Leu Pro Pro Phe Pro Pro			
385	390	395	400
Cys Asn Leu Arg Thr Ala Leu Thr Arg Tyr Ala Cys Leu Leu Ser Ser			
	405	410	415
Pro Lys Lys Ser Ala Leu Val Ala Leu Ala Ala His Ala Ser Asp Pro			
	420	425	430
Thr Glu Ala Glu Arg Leu Lys His Leu Ala Ser Pro Ala Gly Lys Val			
	435	440	445
Asp Glu Tyr Ser Lys Trp Val Val Glu Ser Gln Arg Ser Leu Leu Glu			
	450	455	460
Val Met Ala Glu Phe Pro Ser Ala Lys Pro Pro Leu Gly Val Phe Phe			
465	470	475	480
Ala Gly Val Ala Pro Arg Leu Gln Pro Arg Phe Tyr Ser Ile Ser Ser			
	485	490	495
Ser Pro Lys Ile Ala Glu Thr Arg Ile His Val Thr Cys Ala Leu Val			
	500	505	510
Tyr Glu Lys Met Pro Thr Gly Arg Ile His Lys Gly Val Cys Ser Thr			
	515	520	525
Trp Met Lys Asn Ala Val Pro Tyr Glu Lys Ser Glu Asn Cys Ser Ser			
	530	535	540
Ala Pro Ile Phe Val Arg Gln Ser Asn Phe Lys Leu Pro Ser Asp Ser			
545	550	555	560
Lys Val Pro Ile Ile Met Ile Gly Pro Gly Thr Gly Leu Ala Pro Phe			
	565	570	575
Arg Gly Phe Leu Gln Glu Arg Leu Ala Leu Val Glu Ser Gly Val Glu			
	580	585	590
Leu Gly Pro Ser Val Leu Phe Phe Gly Cys Arg Asn Arg Arg Met Asp			
	595	600	605
Phe Ile Tyr Glu Glu Glu Leu Gln Arg Phe Val Glu Ser Gly Ala Leu			
	610	615	620
Ala Glu Leu Ser Val Ala Phe Ser Arg Glu Gly Pro Thr Lys Glu Tyr			
625	630	635	640
Val Gln His Lys Met Met Asp Lys Ala Ser Asp Ile Trp Asn Met Ile			
	645	650	655
Ser Gln Gly Ala Tyr Leu Tyr Val Cys Gly Asp Ala Lys Gly Met Ala			
	660	665	670
Arg Asp Val His Arg Ser Leu His Thr Ile Ala Gln Glu Gln Gly Ser			
	675	680	685
Met Asp Ser Thr Lys Ala Glu Gly Phe Val Lys Asn Leu Gln Thr Ser			
	690	695	700
Gly Arg Tyr Leu Arg Asp Val Trp			
705	710		

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<211> LENGTH: 641
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AtCPR mutant trAtCPR

<400> SEQUENCE: 8

Met Arg Arg Ser Gly Ser Gly Asn Ser Lys Arg Val Glu Pro Leu Lys
1          5          10          15

Pro Leu Val Ile Lys Pro Arg Glu Glu Glu Ile Asp Asp Gly Arg Lys
          20          25          30

Lys Val Thr Ile Phe Phe Gly Thr Gln Thr Gly Thr Ala Glu Gly Phe
          35          40          45

Ala Lys Ala Leu Gly Glu Glu Ala Lys Ala Arg Tyr Glu Lys Thr Arg
          50          55          60

Phe Lys Ile Val Asp Leu Asp Asp Tyr Ala Ala Asp Asp Asp Glu Tyr
          65          70          75          80

Glu Glu Lys Leu Lys Lys Glu Asp Val Ala Phe Phe Phe Leu Ala Thr
          85          90          95

Tyr Gly Asp Gly Glu Pro Thr Asp Asn Ala Ala Arg Phe Tyr Lys Trp
          100         105         110

Phe Thr Glu Gly Asn Asp Arg Gly Glu Trp Leu Lys Asn Leu Lys Tyr
          115         120         125

Gly Val Phe Gly Leu Gly Asn Arg Gln Tyr Glu His Phe Asn Lys Val
          130         135         140

Ala Lys Val Val Asp Asp Ile Leu Val Glu Gln Gly Ala Gln Arg Leu
          145         150         155         160

Val Gln Val Gly Leu Gly Asp Asp Asp Gln Cys Ile Glu Asp Asp Phe
          165         170         175

Thr Ala Trp Arg Glu Ala Leu Trp Pro Glu Leu Asp Thr Ile Leu Arg
          180         185         190

Glu Glu Gly Asp Thr Ala Val Ala Thr Pro Tyr Thr Ala Ala Val Leu
          195         200         205

Glu Tyr Arg Val Ser Ile His Asp Ser Glu Asp Ala Lys Phe Asn Asp
          210         215         220

Ile Asn Met Ala Asn Gly Asn Gly Tyr Thr Val Phe Asp Ala Gln His
          225         230         235         240

Pro Tyr Lys Ala Asn Val Ala Val Lys Arg Glu Leu His Thr Pro Glu
          245         250         255

Ser Asp Arg Ser Cys Ile His Leu Glu Phe Asp Ile Ala Gly Ser Gly
          260         265         270

Leu Thr Tyr Glu Thr Gly Asp His Val Gly Val Leu Cys Asp Asn Leu
          275         280         285

Ser Glu Thr Val Asp Glu Ala Leu Arg Leu Leu Asp Met Ser Pro Asp
          290         295         300

Thr Tyr Phe Ser Leu His Ala Glu Lys Glu Asp Gly Thr Pro Ile Ser
          305         310         315         320

Ser Ser Leu Pro Pro Pro Phe Pro Pro Cys Asn Leu Arg Thr Ala Leu
          325         330         335

Thr Arg Tyr Ala Cys Leu Leu Ser Ser Pro Lys Lys Ser Ala Leu Val
          340         345         350

Ala Leu Ala Ala His Ala Ser Asp Pro Thr Glu Ala Glu Arg Leu Lys
          355         360         365

His Leu Ala Ser Pro Ala Gly Lys Val Asp Glu Tyr Ser Lys Trp Val
          370         375         380

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Val Glu Ser Gln Arg Ser Leu Leu Glu Val Met Ala Glu Phe Pro Ser
 385 390 395 400
 Ala Lys Pro Pro Leu Gly Val Phe Phe Ala Gly Val Ala Pro Arg Leu
 405 410 415
 Gln Pro Arg Phe Tyr Ser Ile Ser Ser Ser Pro Lys Ile Ala Glu Thr
 420 425 430
 Arg Ile His Val Thr Cys Ala Leu Val Tyr Glu Lys Met Pro Thr Gly
 435 440 445
 Arg Ile His Lys Gly Val Cys Ser Thr Trp Met Lys Asn Ala Val Pro
 450 455 460
 Tyr Glu Lys Ser Glu Asn Cys Ser Ser Ala Pro Ile Phe Val Arg Gln
 465 470 475 480
 Ser Asn Phe Lys Leu Pro Ser Asp Ser Lys Val Pro Ile Ile Met Ile
 485 490 495
 Gly Pro Gly Thr Gly Leu Ala Pro Phe Arg Gly Phe Leu Gln Glu Arg
 500 505 510
 Leu Ala Leu Val Glu Ser Gly Val Glu Leu Gly Pro Ser Val Leu Phe
 515 520 525
 Phe Gly Cys Arg Asn Arg Arg Met Asp Phe Ile Tyr Glu Glu Glu Leu
 530 535 540
 Gln Arg Phe Val Glu Ser Gly Ala Leu Ala Glu Leu Ser Val Ala Phe
 545 550 555 560
 Ser Arg Glu Gly Pro Thr Lys Glu Tyr Val Gln His Lys Met Met Asp
 565 570 575
 Lys Ala Ser Asp Ile Trp Asn Met Ile Ser Gln Gly Ala Tyr Leu Tyr
 580 585 590
 Val Cys Gly Asp Ala Lys Gly Met Ala Arg Asp Val His Arg Ser Leu
 595 600 605
 His Thr Ile Ala Gln Glu Gln Gly Ser Met Asp Ser Thr Lys Ala Glu
 610 615 620
 Gly Phe Val Lys Asn Leu Gln Thr Ser Gly Arg Tyr Leu Arg Asp Val
 625 630 635 640

Trp

<210> SEQ ID NO 9
 <211> LENGTH: 25
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 9

tataccatgg aactgagcag tgtga

25

<210> SEQ ID NO 10
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 10

ctcgaattcg gatccactag tttaataataa agtcgg

36

<210> SEQ ID NO 11
 <211> LENGTH: 43
 <212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 11

ctttaagaag gagatatacc atggcgatgc cgaagaaaag ctc 43

<210> SEQ ID NO 12
 <211> LENGTH: 63
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 12

ctttaagaag gagatatacc atggctctgt tattagcagt tttatgccg aagaaaagct 60

ctt 63

<210> SEQ ID NO 13
 <211> LENGTH: 39
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 13

ctttaagaag gagatatacc atggctaaaa tcgaagaag 39

<210> SEQ ID NO 14
 <211> LENGTH: 35
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 14

ctgaaagacg cgcagactat gccgaagaaa agctc 35

<210> SEQ ID NO 15
 <211> LENGTH: 35
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 15

gagcttttct tcggcatagt ctgcgcgtct ttcag 35

<210> SEQ ID NO 16
 <211> LENGTH: 87
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 16

ctttaagaag gagatatacc atggctaaga aaacgagctc taaagggaag ctcccaccag 60

gacctagcat gccgaagaaa agctctt 87

<210> SEQ ID NO 17
 <211> LENGTH: 44
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

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<400> SEQUENCE: 17
ctttaagaag gagatatacc atggcggact cagaagtcaa tctt 44

<210> SEQ ID NO 18
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 18
gagaacagat tgggtggtatg ccgaagaaaa gctctt 36

<210> SEQ ID NO 19
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 19
aagagctttt ctctggcata ccaccaatct gttctc 36

<210> SEQ ID NO 20
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 20
tataccatgg gtgactgcgt tgccccg 27

<210> SEQ ID NO 21
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 21
cgggatcctt acttcggcag gtcgcgctc 30

<210> SEQ ID NO 22
<211> LENGTH: 29
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 22
cgggatccct tatgcgactc ctgcattag 29

<210> SEQ ID NO 23
<211> LENGTH: 26
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 23
gcccaagctt ttatgccagc atcttc 26

<210> SEQ ID NO 24

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<211> LENGTH: 31
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 24
agatatacat atggttacgg tggaagaata c                               31

<210> SEQ ID NO 25
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 25
ccgctcgagt taggtagcca cactatgcag                               30

<210> SEQ ID NO 26
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 26
ccgctcgagc tagaaataat ttgtttaac                               30

<210> SEQ ID NO 27
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 27
gagcctaggt tagttaccga ttttaaag                               28

<210> SEQ ID NO 28
<211> LENGTH: 38
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 28
gaagatgctg gcataaaagc ttcgatcccg cgaaatta                     38

<210> SEQ ID NO 29
<211> LENGTH: 38
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 29
cgacttaagc attatgcggc cgctacgcc aggttttc                     38

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We claim:

1. A method of producing baicalein and scutellarein, comprising:

- (1) introducing genes expressing flavone 6-hydroxylase and cytochrome P450 oxidoreductase into a host cell to obtain a recombinant strain, wherein said flavone 6-hydroxylase is a mutant flavone 6-hydroxylase with the

60

N-terminal amino acids 2 to 25 truncated and fused with a peptide tag, said peptide tag is a 2B1 family soluble protein of cytochrome P450, and said host cell is a *Escherichia coli* cell;

- (2) culturing the recombinant host cell in a culture system containing chrysin or apigenin to produce baicalein or scutellarein.

65

49

2. A method of producing baicalein and scutellarein, comprising:

(1) introducing genes expressing flavone 6-hydroxylase and cytochrome P450 oxidoreductase, as well as genes for synthesizing chrysin or apigenin, into a host cell, wherein said flavone 6-hydroxylase is a mutant flavone 6-hydroxylase with the N-terminal amino acids 2 to 25 truncated and fused with a peptide tag, said peptide tag is a 2B1 family soluble protein of cytochrome P450, and said host cell is a *Escherichia coli* cell;

(2) culturing the host cell in a culture system containing phenylalanine and/or tyrosine to produce baicalein or scutellarein.

3. The method according to claim 2, wherein, the genes for synthesizing chrysin or apigenin comprises: genes expressing phenylalanine ammonia-lyase, 4-coumarate: CoA ligase, chalcone synthase, chalcone isomerase and flavone synthase I.

50

4. The method according to claim 3, wherein, when introduced into the host cell, the genes expressing phenylalanine ammonia-lyase, 4-coumarate: CoA ligase, chalcone synthase, chalcone isomerase and flavone synthase I are in the same expression vector.

5. The method according to claim 2, wherein, the cytochrome P450 oxidoreductase is a full-length or mutant cytochrome P450 oxidoreductase with the N-terminal amino acids (1-20) to (60-85) truncated.

6. The method according to claim 2, wherein, the cytochrome P450 oxidoreductase is a full-length or mutant cytochrome P450 oxidoreductase with the N-terminal amino acids (2-10) to (65-80) truncated.

7. The method according to claim 2, wherein, the cytochrome P450 oxidoreductase is a full-length or mutant cytochrome P450 oxidoreductase with the N-terminal amino acids (2-5) to (70-75) truncated.

* * * * *